Carbon and Its Compounds

❖ Carbon

- ✓ It is a non-metallic Element.
- ✓ The name 'carbon' comes from the Latin word **'carbo'**, meaning **coal**, because carbon is the main constituent of coal.
- ✓ Its atomic number is 6.
- ✓ Its valency is 4.
- ✓ The four valencies of carbon are shown by four short lines (bonds) drawn around the symbol C – one on each side.
- ✓ Carbon has 4 valence electrons, allowing it to form four covalent bonds with other atoms.
- ✓ It is first element in group 14 of Modern periodic table.
- ✓ **Organic chemistry** is a branch of chemistry based on carbon and its compounds.
- ✓ Carbon forms **millions of compounds**, more than any other element
- ✓ Almost everything around us that is not metallic and not made up of glass or clay contains carbon.
- ✓ Things made of carbon compounds in daily life:
  - © **Food materials:** grains, pulses, sugar, tea, coffee, fruits and vegetables.
  - © **Clothing materials:** cotton, silk, wool, nylon and polyester.
  - © **Fuels:** wood, coal, kerosene, LPG, natural gas, CNG, petrol and diesel.
  - © **Others:** paper, rubber, plastics, leather, drugs (medicines) and dyes.
- ✓ **Occurrence of Carbon in Nature**
  - © **Free (elemental) state:** mainly as diamond and graphite; a third form, buckminsterfullerene, also occurs naturally. Only a small amount of carbon occurs in the free state.
  - © **Combined state:** most carbon occurs as compounds – carbon dioxide gas in air, carbonates, fossil fuels and organic compounds.
  - © Found in **living organisms** as organic compounds such as carbohydrates, fats and proteins, and in materials obtained from them like cotton, wool and wood.
  - © Present in **fossil fuels** (coal, petroleum, natural gas).
  - © Exists as **carbonates** in minerals (limestone, marble and chalk).
- ✓ **Earth's crust has** Only **0.02%** carbon (as carbonates, coal, petroleum, etc.).
- ✓ **Atmosphere** Only **0.03%** carbon in the form of carbon dioxide (textbook value; the present-day level is about 0.04%).
- ✓ **Biological significance of carbon**
  - © Almost all biomolecules (carbohydrates, proteins, lipids, nucleic acids) are made up of Carbon.
  - © It is Essential for photosynthesis (CO_2) and respiration (glucose).
- ✓ **Industrial & Technological significance of carbon**
  - © Carbon is used in fuels, polymers, medicines, dyes, and nanomaterials Industry.
  - © Carbon is key component in steel production.
- ✓ **Test for Carbon Compounds: Combustion Test**
  - © Burn a small sample of the item to be tested(in safe environment) and observe:
  - © If the item chars or burns leaving a black, sooty deposit, it indicates the presence of carbon. Note: saturated compounds burn with a clean blue flame and leave no soot, so the absence of soot does not prove that carbon is absent – confirm with the limewater test below.
  - © The burning of carbon-containing compounds usually produces carbon dioxide (CO_2).
- ✓ **Test for Carbon Dioxide (CO_2): Limewater Test**
  - © Pass the gas produced from burning through limewater, $Ca(OH)_2$ solution.

© If the limewater turns milky, it confirms the presence of carbon dioxide, and hence the compound contains carbon.

❖ Carbon Compounds and Covalent Bonding

- ✓ Carbon compounds, also known as organic compounds, are chemical substances that contain carbon atoms bonded to other elements such as hydrogen, oxygen, nitrogen, Sulphur, or halogens.
- ✓ They are essential to life, as they make up the molecular structure of living organisms.
- ✓ Carbon's ability to form stable, covalent bonds with itself and other elements allows for a vast diversity of organic compounds, ranging from simple molecules like methane (CH_4) to complex biomolecules like proteins and DNA.
- ✓ Most carbon compounds are poor conductors of electricity.
- ✓ These compounds generally have low melting and boiling points, unlike ionic compounds.
- ✓ **Carbon has four valence electrons (atomic number 6, configuration 2, 4). To achieve a stable configuration, it would need to gain or lose four electrons. Gaining four electrons to form C^{4-} is difficult due to repulsion between 10 electrons and only 6 protons. Losing four electrons to form C^{4+} requires too much energy and results in an unstable ion. Hence, carbon forms covalent bonds by sharing electrons instead of forming ionic bonds.**
- ✓ Carbon atom is very small, so its outermost electrons are held strongly by the nucleus – that is another reason why it cannot lose 4 electrons easily. Since carbon can attain the inert-gas configuration only by sharing electrons, **carbon always forms covalent bonds**.
- ✓ **Bonding in Carbon Compounds – Covalent Bonding**
  - © Covalent bonding is a type of chemical bond formed by the sharing of electrons between atoms.
  - © It is formed between nonmetals or metalloids (e.g., C, O, N, H).
  - © Covalent Bonding is Common in organic compounds (e.g., hydrocarbons, proteins) due to carbon's ability to share electrons and form diverse structures.
  - © **Not only carbon**, but many other elements also form stable molecules through **covalent bonding** by **sharing electrons**, rather than gaining or losing them.

Difference between Covalent and Ionic Bonding	
Covalent Bonding	**Ionic Bonding**
Electrons are shared between atoms.	Electrons are transferred (from metal to non-metal).
Forms between two non-metals (or metalloids).	Forms between metals and non-metals.
Low melting/boiling points, because the forces between molecules (intermolecular forces) are weak.	High melting/boiling points, because of the strong electrostatic attraction between oppositely charged ions.
Poor conductors of electricity (no charged particles).	Conduct electricity when molten or dissolved in water (mobile ions).
No ions are formed; separate molecules are formed.	Ions (Na^+, Cl^-) are formed; they form a crystal lattice, not separate molecules.
Generally soft or flexible (e.g., plastics, wax).	Generally hard and brittle.
Usually soluble in organic solvents (e.g., ethanol); many are insoluble in water.	Usually soluble in water; insoluble in organic solvents like kerosene and petrol.
Generally gases, liquids or low-melting solids at room temperature.	Solids at room temperature.
Reactions are usually slow, because covalent bonds must break.	Reactions in solution are usually fast, because free ions react directly.
Examples: Methane (CH_4), Sugar ($C_{12}H_{22}O_{11}$).	Examples: Common salt (NaCl), Baking soda ($NaHCO_3$).

❖ Versatile nature of Carbon

✓ Chemists have identified millions of carbon compounds. This number is more than the total compounds formed by all other elements combined. Carbon has two special properties that allow it to form countless compounds:

✓ More than **5 million** carbon compounds are known today, and many more are being prepared in laboratories every day.

1) Catenation (Self-Linking Ability)

© Catenation is the ability of an element to form bonds with atoms of the same element to make long chains, branched chains, or rings.

© Carbon atoms can join with other carbon atoms to form long chains and thus it shows catenation.

© Examples of long chains: **octane (C_8H_{18})**, a constituent of petrol, has a chain of 8 carbon atoms; **starch and cellulose** contain chains of hundreds of carbon atoms.

© The long chain of carbon atoms acts as a **backbone** to which other atoms can attach in many ways – this gives an extremely large number of carbon compounds.

© Carbon chains can be:

- Straight (unbranched)
- Branched
- Ring-shaped

© Carbon atoms in these chains can be connected by:

- Single bonds**(C–C)**: Saturated compounds
- Double **(C=C)** or triple bonds **(C≡C)**: Unsaturated compounds

© No other element shows catenation as strongly as carbon.

© Other elements which show catenation (but to a lesser extent than Carbon) are:

- Silicon (Si) – forms chains of up to 7–8 atoms
- These silicon–hydrogen compounds are very reactive and unstable because the Si–Si bonds are weak.
- Sulphur (S) – forms rings (e.g., S_8)
- Phosphorus (P) – forms comparatively less stable chains in some compounds.

2) Tetravalency – Carbon's 4-Bonding Capacity

© Carbon has a valency of 4, meaning it can bond with:

- **4 other carbon atoms** (forming complex structures).
- **Other elements** like hydrogen (H), oxygen (O), nitrogen (N), sulphur (S), chlorine (Cl), etc.

© These combinations create millions of different compounds with unique properties.

✓ Strong covalent bond and stable compounds formation by carbon:

© Carbon forms strong covalent bonds due to its small atomic size.

© Carbon is a small atom, meaning its nucleus is close to the shared electrons in a bond. This allows the nucleus to hold onto the electrons tightly, making the bond strong and stable.

© Larger atoms (like silicon) form weaker bonds because their nuclei can't hold electrons as strongly.

❖ Electron Dot Structure (Lewis Dot Structure)

© An electron dot structure is a simple way to represent valence electrons (outermost electrons) of atoms using dots or crosses around the chemical symbol.

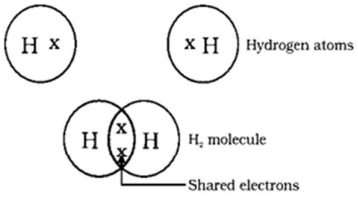
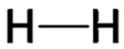
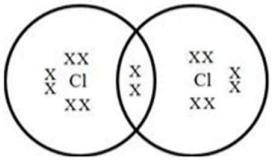
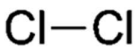
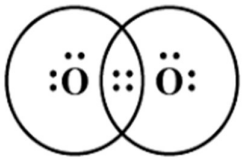
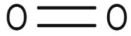
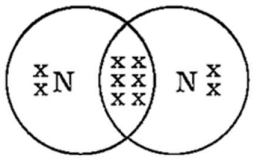
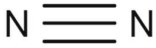
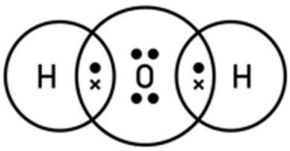
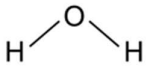
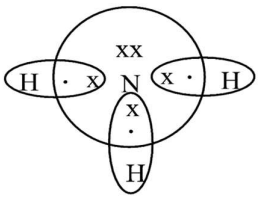
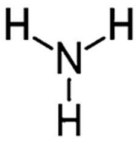
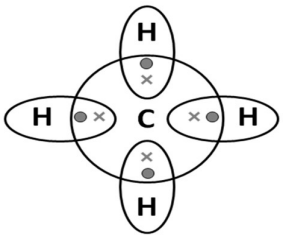
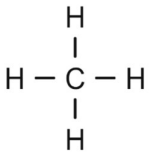
© It shows how atoms share or transfer electrons to form bonds (ionic or covalent).

© It helps us understand the bonding and stability of molecules.

© Important points while making Electron dot structure:

- **Only Valence Electrons are Shown**: We do not show inner electrons. Only electrons in the outermost shell (valence shell) are represented.
- **Dots/Cross Represent Electrons**: Each dot or cross stands for one valence electron. Electrons from different atoms may be shown using different symbols (like one with dots, one with crosses).

- **Covalent Bonding in Dot Structure:** When atoms share electrons, the shared pair is placed between the symbols of the atoms. One shared pair is equal to one covalent bond.

Molecule	Atoms Involved	Bond Type	Electron Dot Structure	Explanation	Structure
Hydrogen (H_2)	2 Hydrogen (H) atoms	Single Covalent		Each hydrogen shares 1 electron to complete 2 electrons (duplet)	
Chlorine (Cl_2)	2 Chlorine (Cl) atoms	Single Covalent		Each chlorine shares 1 electron to complete 8 electrons (octet)	
Oxygen (O_2)	2 Oxygen (O) atoms	Double Covalent		Each oxygen shares 2 electrons to complete octet	
Nitrogen (N_2)	2 Nitrogen (N) atoms	Triple Covalent		Each nitrogen shares 3 electrons to complete octet	
Water (H_2O)	1 Oxygen, 2 Hydrogen atoms	2 Single Covalent		Oxygen shares 1 electron with each hydrogen atom	
Ammonia (NH_3)	1 Nitrogen, 3 Hydrogen atoms	3 Single Covalent		Nitrogen shares 1 electron each with 3 hydrogen atoms	
Methane (CH_4)	1 Carbon, 4 Hydrogen atoms	4 Single Covalent		Carbon shares 1 electron each with 4 hydrogen atoms	

❖ Electron Dot Structures of Carbon Compounds

✓ **Steps to draw the structure of a carbon compound**

© **Step 1:** Link the carbon atoms together with single bonds (e.g., C–C for ethane).

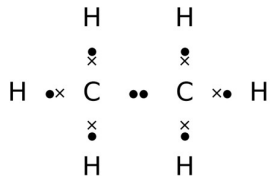
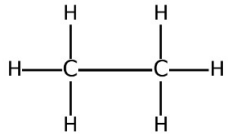
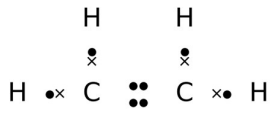
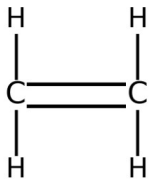
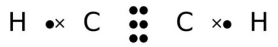
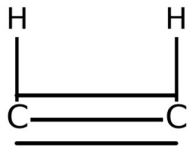
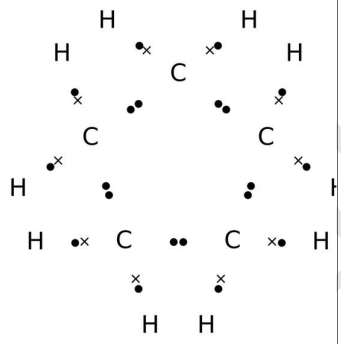
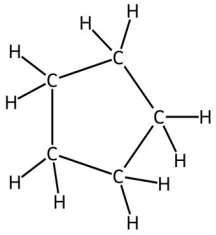
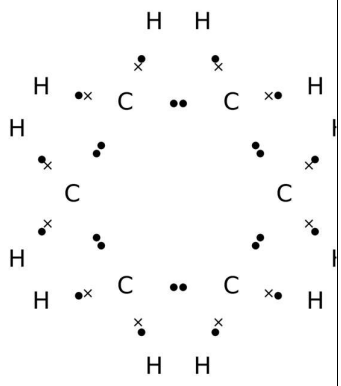
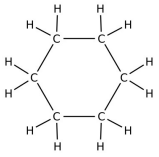
© **Step 2:** Use hydrogen atoms to satisfy the remaining valencies of each carbon atom.

© **Step 3:** If a valency per carbon still remains unsatisfied (as in C_2H_4 or C_2H_2), it can be satisfied only by a double or triple bond between the carbon atoms.

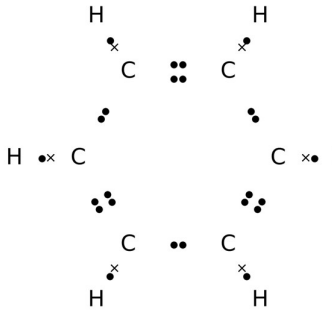
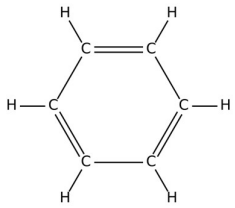
✓ **Representing Bonds in Electron Dot Structures**

© A **single bond** (one shared pair) is shown by **2 dots/crosses**, a **double bond** by **4**, and a **triple bond** by **6** between the two atoms.

© In the structures below, in each C–H bond the electron of carbon is shown as a cross (×) and the electron of hydrogen as a dot (•); electrons shared between two carbon atoms are shown as dots.

Molecule	Electron Dot Structure	Structural Formula	Bonds present
Ethane C_2H_6 (alkane)			1 C–C single bond + 6 C–H single bonds = **7 covalent bonds**
Ethene C_2H_4 (alkene)			1 C=C double bond (2 shared pairs) + 4 C–H single bonds
Ethyne C_2H_2 (alkyne)			1 C≡C triple bond (3 shared pairs) + 2 C–H single bonds
Cyclopentane C_5H_{10}			5 C–C + 10 C–H single bonds = **15 covalent bonds**
Cyclohexane C_6H_{12}			6 C–C + 12 C–H single bonds = **18 covalent bonds**

Conceptual Learning

Molecule	Electron Dot Structure	Structural Formula	Bonds present
Benzene C_6H_6 (unsaturated, cyclic)			3 C=C double bonds + 3 C–C single bonds (alternate) + 6 C–H single bonds

- ✓ **Condensed Formulae of Hydrocarbons**
  - © Ethane is written as **$H_3C–CH_3$** or simply **$CH_3–CH_3$**; ethene as **$H_2C=CH_2$** or **$CH_2=CH_2$**; ethyne as **HC≡CH** or **CH≡CH**.
  - © It is always understood that the single, double or triple bond is **between the carbon atoms** (not between carbon and hydrogen).

❖ Carbon Allotropes

- ✓ Allotropes are different structural forms of the same chemical element, where the atoms are bonded together in distinct arrangements. These variations result in different physical properties (such as hardness, colour, conductivity) while being made of the same element. Their chemical properties are generally similar, but not always identical – e.g., ozone (O_3) is far more reactive than oxygen (O_2), and white phosphorus is far more reactive than red phosphorus.
- ✓ Carbon exists in multiple forms (allotropes) with similar chemical properties (all of them burn in oxygen to give CO_2) but different physical properties.
- ✓ The main allotropes of Carbon are **Diamond, Graphite and Fullerenes** (graphene and carbon nanotubes are other forms).
- ✓ Differences between allotropes of Carbon arise from variation in carbon-carbon bonding patterns.
- ✓ Examples of elements other than Carbon which have allotropes are:
  - © **Oxygen:** O_2 (diatomic oxygen) and O_3 (ozone)
  - © **Phosphorus:** White Phosphorus(P_4), Red phosphorus(polymeric Chain of P atoms) and Black phosphorus (layered sheet structure of Phosphorus atoms.)
  - © **Sulphur:** Rhombic Sulphur and monoclinic Sulphur
- ✓ **Diamond**

⇒ **Structure**

- © Each carbon covalently bonded to 4 other carbons.
- © Forms rigid 3D tetrahedral network.
- © Strong covalent bonds in all directions.
- © A diamond crystal is a **giant molecule** of carbon atoms; the four surrounding carbon atoms lie at the four corners of a **regular tetrahedron**.
- © All 4 valence electrons of each carbon are used in bonding, so there are **no free electrons**.

⇒ **Appearance and other properties:**

- © Diamond is a **colourless, transparent** substance with **extraordinary brilliance**.
- © It is quite heavy – it has a **high density** (about 3.5 g/cm^3) because of its compact, rigid 3D structure.
- © Its melting point is very high – **more than 3500 °C** – because a lot of energy is needed to break the network of strong covalent bonds.
- © Diamond burns on strong heating in oxygen to form **only carbon dioxide**, leaving nothing behind (the CO_2 turns lime water milky). This proves diamond is made of carbon only, so its symbol is C.

⇒ **Properties of diamond:**

Property	Reason
Hardness(Hardest known substance)	3D tetrahedral covalent network
High melting point	Strong covalent bonds require massive energy to break
Electrical insulator	No free electrons; all electrons locked in bonds
Chemically inert	No reactive sites; stable C–C bonds
High refractive index	Dense atomic structure slows light

⇒ **Synthetic Diamond:**

© Synthetic diamonds can be made by subjecting pure carbon to very high pressure and temperature.

© These synthetic diamonds are small but have identical properties to natural diamonds.

© Synthetic diamonds are usually not good enough for jewellery; they are used in glass cutters, diamond-studded saws and drill bits.

⇒ **Uses of Diamond:**

© Being extremely hard, diamond is used for **cutting and grinding hard materials** – in **glass cutters**, saws for cutting **marble**, and **rock-drilling equipment** (for drilling oil wells).

© **Diamond dies** are used for drawing thin wires, such as the tungsten filament of an electric bulb.

© Diamonds are used for making **jewellery** because of their extraordinary brilliance, which is due to their great ability to **reflect and refract light**.

© Sharp-edged diamond knives are used by **eye surgeons** to remove cataract from the eyes.

✓ **Graphite**

⇒ **Structure:**

© Each carbon atom is bonded to three other carbon atoms in the same plane.

© Graphite is a **greyish-black, opaque** substance; it is **lighter than diamond**.

© Forms 2D hexagonal layers or hexagonal arrays.

© One of the bonds is a double bond, satisfying carbon's valency.

© These hexagonal sheets are stacked in layers.

© Layers are held together by weak van der Waals forces, allowing them to slide over each other.

© The layers are quite far apart, so **no covalent bonds** exist between them.

© Each carbon uses only 3 of its 4 valence electrons in bonding; the **fourth electron is free** to move – this makes graphite a good conductor of electricity.

© Graphite also burns in oxygen to give **only carbon dioxide**, proving it is made of carbon only.

⇒ **Physical Properties:**

Physical Property	Reason
Soft and slippery	Weak van der Waals forces between layers allow the layers to slide over each other.
Good electrical conductor	Each carbon uses only 3 of its 4 valence electrons in bonding; the remaining delocalised electrons move freely along (within) the layers.
Opaque, greyish-black and shiny	Delocalised electrons absorb visible light.
Good thermal conductor	Delocalised electrons carry heat efficiently along the layers.
Lower density than diamond (≈2.2 vs 3.5 g/cm³)	Large spacing between the hexagonal layers.
Very high melting point (~3,600 °C; sublimes)	Strong covalent bonds within the layers need a lot of energy to break.
Fairly unreactive	Stable covalent bonds within the layers; slightly more reactive than diamond, mainly at the layer edges.

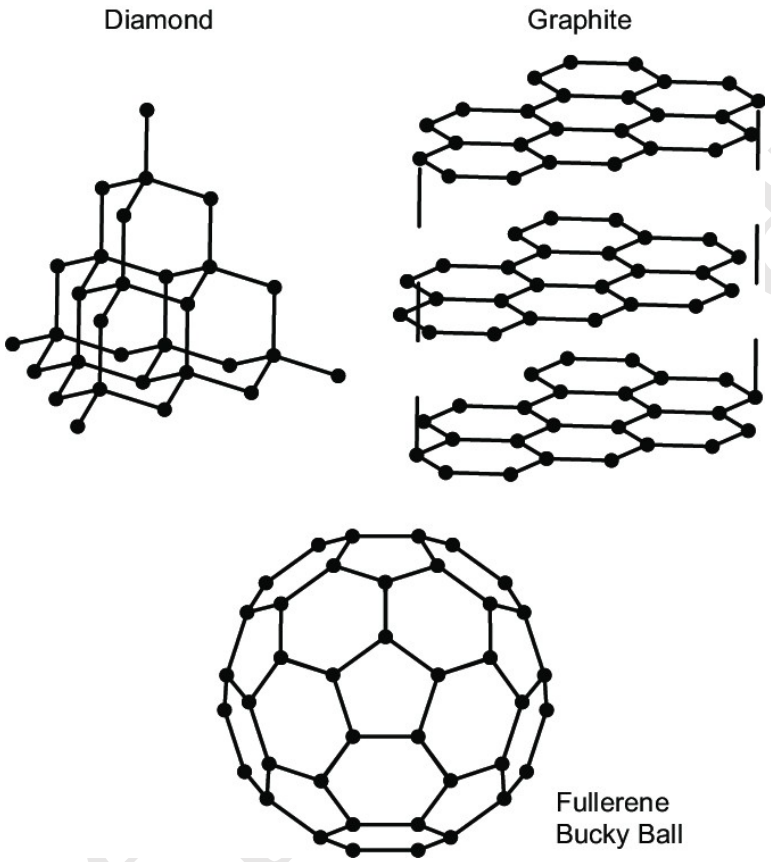
⇒ **Uses of Graphite:**

© **Lubricant:** powdered graphite is used as a dry lubricant for fast-moving machine parts. Being non-volatile, it can lubricate parts working at **very high temperatures** where oil cannot be used. It is used as powder, mixed with petroleum jelly (graphite grease) or mixed with lubricant oils.

© **Electrodes:** being a good conductor, graphite is used to make electrodes in **dry cells** (the black 'anode' rod) and in **electric arcs**; carbon brushes of **electric motors** are also made of graphite.

© **Pencil leads:** graphite is black and quite soft, so it marks black lines on paper. Mixed with clay, it is used to make pencil 'leads' (graphite is also called **black lead**). It is also used in **black paints**.

© Since graphite does not melt easily, it is used in the tiles on the **nose cone of a space shuttle**.



✓ **Fullerenes:**

⇒ Fullerenes are a new class of carbon allotropes.

⇒ C_{60} is named Buckminsterfullerene after the architect Buckminster Fuller, who designed geodesic domes with a similar structure.

⇒ It is used in nanotechnology, lubricants, biomedical device and cosmetics.

⇒ **Structure of Fullerenes:**

© The most famous and first discovered fullerene is C_{60}, which resembles a football. C_{70} is another type of fullerene.

© The C_{60} molecule has 60 carbon atoms arranged in a soccer ball or geodesic dome with 20 hexagons and 12 pentagons.

© The molecule is made of **interlocking hexagonal and pentagonal rings** of carbon atoms, like a football with each corner of a panel being a carbon atom.

⇒ **Properties of Buckminsterfullerene:**

© It is a **dark solid** at room temperature and occurs naturally in small amounts.

© Diamond and graphite are **giant molecules** (unending networks of atoms), but buckminsterfullerene is a **small, separate molecule** of only 60 carbon atoms.

© It is **neither very hard nor soft** (diamond is extremely hard, graphite is soft). Its other properties are still being investigated.

© It also burns in oxygen to give only carbon dioxide, showing that it is made of carbon only.

⇒ **Difference between Diamond and Graphite:**

Property	Diamond	Graphite
Structure	Each C joined to 4 other C atoms; rigid 3D tetrahedral network	Each C joined to 3 other C atoms; flat hexagonal layers held by weak forces
Appearance	Colourless, transparent, brilliant	Greyish-black, opaque
Hardness	Hardest natural substance	Soft and slippery
Density	High (about 3.5 g/cm³)	Lower (about 2.2 g/cm³)
Electrical conductivity	Non-conductor (no free electrons)	Good conductor (one free electron per C atom)
Uses	Cutting tools, rock drills, jewellery	Lubricant, electrodes, pencil leads
Chemical property	Burns in oxygen to give only CO_2	Burns in oxygen to give only CO_2 (same chemical properties)

❖ History of Carbon Compounds

⇒ Historically, carbon-based (organic) compounds were initially extracted only from natural, biological substances.

⇒ The Vital Force Theory: Because of this natural association, it was believed that organic compounds could only be synthesized within living systems, leading to the postulate that a "vital force" was mandatory for their creation.

⇒ The Theory Disproved (1828): **Friedrich Wöhler** disproved this theory in 1828 by successfully synthesizing urea (an organic compound) from ammonium cyanate (an inorganic compound) in a laboratory setting.

$$NH_4CNO \xrightarrow{\Delta} NH_2CONH_2$$

$$(\text{Ammonium Cyanate}) \xrightarrow{\text{Heat}} (\text{Urea})$$

⇒ **Scope of Organic Chemistry:** Even after Wöhler's discovery, not every carbon compound is classed as organic. Carbides, oxides of carbon (CO, CO_2), carbonates and hydrogencarbonates are studied under inorganic chemistry, because they lack Carbon–Hydrogen (C-H) bonds and behave like simple inorganic (often ionic) compounds rather than typical organic compounds built on carbon chains.

❖ Hydrocarbons

✓ Hydrocarbons are organic compounds composed entirely of carbon and hydrogen atoms.

✓ The most important source of hydrocarbons is **petroleum (crude oil)**, obtained from underground oil deposits by drilling oil wells. **Natural gas**, which occurs above petroleum deposits, also contains hydrocarbons.

✓ Unsaturated hydrocarbons are obtained from petroleum by a process called **cracking**.

✓ They form the parent framework of organic chemistry, since every other class of organic compound can be regarded as a hydrocarbon in which one or more hydrogen atoms are replaced by a functional group.

✓ **Classification of Hydrocarbons:**

1) Aliphatic Hydrocarbons

© Open-chain (straight or branched) compounds, or non-aromatic ring compounds, in which carbon atoms are joined by single, double, or triple bonds.

© There are two types of Aliphatic Hydrocarbons: **Acyclic and Alicyclic Hydrocarbons**

(a) Acyclic Hydrocarbons:

⇒ Hydrocarbons whose carbon atoms are joined in an open chain with or without branches.

⇒ There are two types of Acyclic Hydrocarbons: **Saturated and unsaturated Hydrocarbons**

(b) Alicyclic Hydrocarbons

Concept Learning

⇒ Hydrocarbons whose carbon atoms are joined in a closed (non-aromatic) ring.

⇒ There are two types of Alicyclic Hydrocarbons: **Saturated and unsaturated Hydrocarbons.**

2) Aromatic Hydrocarbons

© Cyclic, planar compounds with a delocalised, conjugated π-electron cloud that satisfies Hückel's rule (4n+2 π-electrons).

© Benzene (C6H6), Toluene (C7H8).

✓ **Homologous Series**

© A homologous series is a family of organic compounds that can all be represented by the same general formula, possess similar chemical properties (because they share the same functional group), and in which each successive member differs from the one before it by a constant unit of $–CH_2–$ (14 atomic mass units).

© Just as elements with similar electron structures are placed in the same group of the periodic table, organic compounds with similar structures and properties are placed in the same homologous series, arranged in order of increasing molecular mass.

© The members of a homologous series are called **homologues**. Two adjacent homologues differ by 1 carbon atom and 2 hydrogen atoms (a $–CH_2–$ group).

© The existence of homologous series has **simplified the study of organic chemistry** – instead of studying millions of compounds separately, we study only a few series.

© **Characteristics of a Homologous Series:**

- Same general formula — e.g. all alkanes fit C_nH_{2n+2} (methane = CH_4 when n=1, ethane = C_2H_6 when n=2).
- Constant CH_2 difference — each member's molecular mass increases by 14 units over the previous one (ethane, 30 amu, is 14 more than methane, 16 amu).
- Similar chemical properties — since all members share the same functional group, they undergo similar characteristic reactions (e.g. primary and secondary alcohols are oxidised to aldehydes/ketones; tertiary alcohols resist mild oxidation).
- Gradually changing physical properties — boiling point, melting point and density increase steadily with molecular mass (methane: –162°C; ethane: –89°C; propane: –42°C).
- Prepared by similar general methods — e.g. all alcohols can be made by hydrolysis of the corresponding haloalkane.

© **Common Homologous Series:**

S.No	Series	General Formula	Functional Group
1	Alkanes	C_nH_{2n+2}	None (C–C, C–H only)
2	Alkenes	C_nH_{2n}	C=C (one double bond)
3	Alkynes	C_nH_{2n-2}	C≡C (one triple bond)
4	Alcohols	$C_nH_{2n+1}OH$	–OH (hydroxyl)
5	Aldehydes	$C_nH_{2n}O$	–CHO
6	Ketones	$C_nH_{2n}O$	C=O (>C=O)
7	Carboxylic Acids	$C_nH_{2n}O_2$	–COOH

✓ **Saturated Hydrocarbons (Alkanes):**

© Hydrocarbons containing only carbon–carbon single (sigma) bonds — the maximum possible number of hydrogen atoms are attached.

© They are called saturated because all carbon atoms are bonded by single bonds, meaning they already hold the maximum possible number of hydrogen atoms without breaking the carbon skeleton.

- © **Alkanes (C_nH_{2n+2})** and **Cycloalkanes (C_nH_{2n})** fall under the saturated category because their carbon backbones consist entirely of single bonds (C-C).

✓ **Unsaturated Hydrocarbons (Alkenes & Alkynes)**

- © Hydrocarbons containing at least one carbon–carbon double bond (alkenes) or triple bond (alkynes), so they have fewer hydrogens than the corresponding saturated compound.
- © They are called unsaturated because they contain double or triple carbon-carbon bonds, allowing them to accept additional hydrogen atoms without breaking the main carbon skeleton.
- © **Alkenes (C_nH_{2n})** and **Alkynes (C_nH_{2n-2})** fall under the unsaturated category because their carbon backbones contain double (C=C) or triple (C ≡ C) bonds.
- © Unsaturated hydrocarbons (alkenes and alkynes) are **chemically more reactive** than saturated hydrocarbons (alkanes), which are quite unreactive.
- © The simplest alkene and alkyne have **2 carbon atoms** (ethene and ethyne) – there can be no alkene or alkyne with only one carbon atom.
- © **Test for unsaturation – Bromine water test:** an unsaturated compound **decolourises** the red-brown colour of bromine water, while a saturated compound does not (details under Addition Reaction).

✓ **Identifying an Alkane, Alkene or Alkyne from its Formula**

- © If the number of H atoms is **2 more than double** the number of C atoms → **alkane** (C_nH_{2n+2}), e.g. C_3H_8.
- © If the number of H atoms is **exactly double** the number of C atoms → **alkene** (C_nH_{2n}), e.g. C_3H_6 (or a cycloalkane).
- © If the number of H atoms is **2 less than double** the number of C atoms → **alkyne** (C_nH_{2n-2}), e.g. C_3H_4.

✓ **Uses of Some Hydrocarbons**

- © **Methane** (natural gas, CNG) is used as a fuel in homes, industry and vehicles; **LPG** is mainly **butane** (a mixture of n-butane and isobutane).
- © **Petrol** is a mixture of alkanes; **candle wax** is a mixture of heavier alkanes (high molecular masses).
- © **Ethene** is used to make **polythene** (carry bags etc.) and for **ripening raw fruits**.
- © **Ethyne (acetylene)** is used in **acetylene lamps** and in oxy-acetylene flames for welding; it also forms a polymer called polyacetylene.

✓ **IUPAC Nomenclature of Hydrocarbons:**

- © **IUPAC = International Union of Pure and Applied Chemistry**
- © The world body that decides official chemical names.
- © Before IUPAC, chemists used trivial/common names (like 'wood spirit' for methanol, 'grape sugar' for glucose). These differed from country to country and caused confusion.
- © 1919: IUPAC was founded, to standardise chemical names worldwide.
- © Since then, IUPAC has kept updating these rules so that one compound = one unambiguous name, understood by any chemist in any country.

✓ **IUPAC Nomenclature of Basic Hydrocarbons:**

- © **Root Names (Carbon Count)**
  - The root tells how many carbons are in the chain.

No of Carbon atoms "C"	Root	No of Carbon atoms "C"	Root
1	Meth	6	Hex
2	Eth	7	Hept
3	Prop	8	Oct
4	But	9	Non
5	Pent	10	Dec

© **Suffix for Bond Type**

Root + Suffix = Name. Example: Eth + ane = Ethane.

Bond Type	Suffix	Example
Only single bonds	**-ane**	**Ethane**
One or more C=C	**-ene**	**Ethene**
One or more C≡C	**-yne**	**Ethyne**

© Rules for Naming Hydrocarbons:

(1) Find the Parent Chain

⇒ Find the longest continuous carbon chain.

⇒ If a ring is present, the ring is usually the parent (add prefix 'cyclo-').

⇒ *Example: In a 5-carbon chain with a branch, the parent is the longest chain, not the branch.*

(2) Number the Chain (Lowest Locants)

⇒ Number from the end gives the lowest locant, in this order: (1) C=C/C≡C bond first, (2) substituents next.

⇒ Example: CH_2=CH–CH_2–CH_3 is But-1-ene, not but-3-ene (lowest number to the double bond). With a substituent: CH_2=CH–CH(CH_3)–CH_3 is 3-methylbut-1-ene, not 2-methylbut-3-ene — the double bond gets priority over the substituent.

(3) Name Substituents as Prefixes

⇒ Alkyl branches and halogens are named as prefixes, each with its own locant.

⇒ Example: Chlorine on C2 of propane → 2-chloropropane.

(4) Alphabetise the Prefixes

⇒ List prefixes in alphabetical order (by spelling), not by locant number or drawing order.

⇒ Example: 4-ethyl-2-methylhexane — 'ethyl' before 'methyl' even though its number is bigger.

(5) Repeated Substituents

⇒ Use di-, tri-, tetra- for repeats.

⇒ Give every occurrence its own locant, separated by commas. Ignore di-/tri- when alphabetising.

⇒ Example: 2,3-Dimethylbutane — comma between the locants, not a hyphen.

(6) Both a Double AND a Triple Bond

⇒ Cite both in one suffix, always 'ene' before 'yne': '-en-yne'. Number for the lowest combined locants; if tied, the double bond wins.

⇒ Example: Pent-1-en-4-yne — double bond at C1, triple bond at C4.

(7) Assemble the Final Name

⇒ Order: alphabetised prefixes (with locants) + parent chain + suffix. Hyphen between a locant and a word; comma between locants.

⇒ Example: 4-Ethyl-2-methylhexane.

✓ **Isomerism:**

© It is the phenomenon in which two or more compounds have the exact same molecular formula but differ in the arrangement of their atoms in space or in structure, causing them to have different physical and/or chemical properties.

© In inorganic chemistry a molecular formula usually represents only one compound (e.g., H_2SO_4 is only sulphuric acid), but in organic chemistry the **same molecular formula can represent two or more compounds**, because the carbon atoms can be arranged in different ways.

© Compounds with the same molecular formula but different structures are called **isomers** (structural isomers), e.g. n-butane and iso-butane (both C_4H_{10}). LPG contains both these isomers.

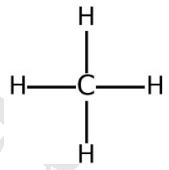
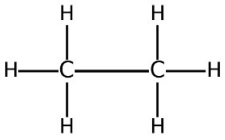
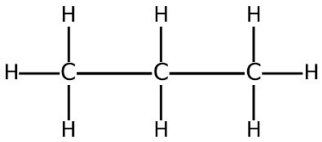
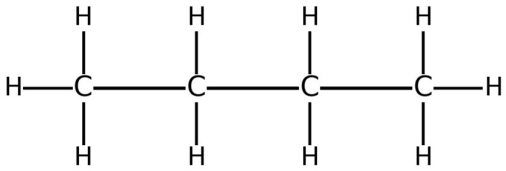
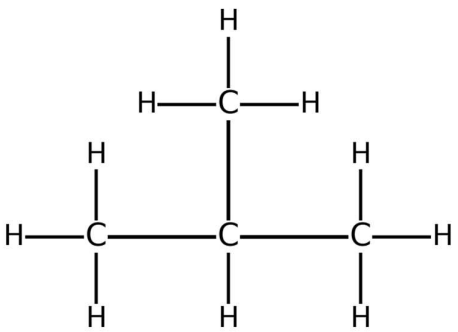
© Isomerism is possible only in hydrocarbons with **4 or more carbon atoms**. Methane, ethane and propane (1, 2 or 3 carbon atoms) show **no isomerism**.

Conceptual Learning

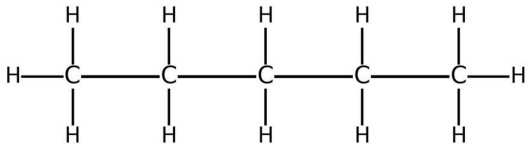
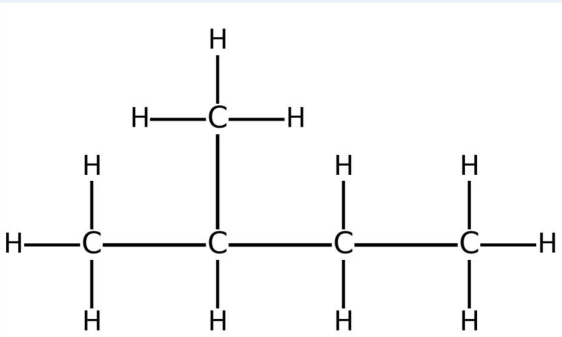
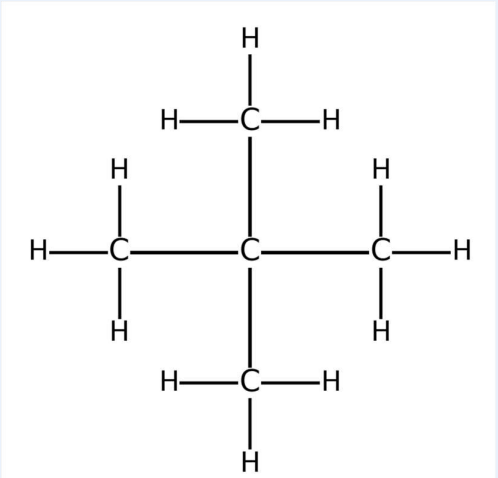
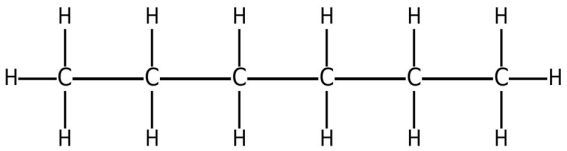
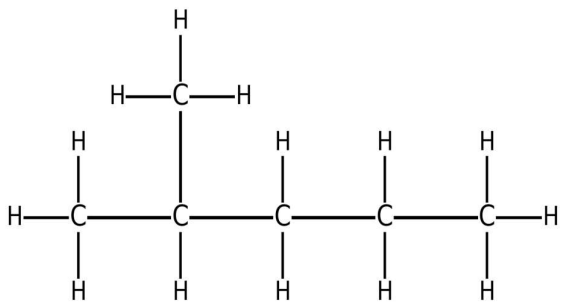
- © Number of isomers: **butane – 2, pentane – 3, hexane – 5**. The number of isomers increases rapidly as the number of carbon atoms increases.
- © To draw all isomers of an alkane: first write all carbon atoms in one straight chain (the **normal** or **n-** isomer), then draw the branched arrangements by shortening the main chain step by step.
- © A straight-chain isomer is called **normal (n-)**, e.g. n-butane, n-pentane. For hydrocarbons with up to 3 carbon atoms, the common and IUPAC names are the same (methane, ethane, propane).

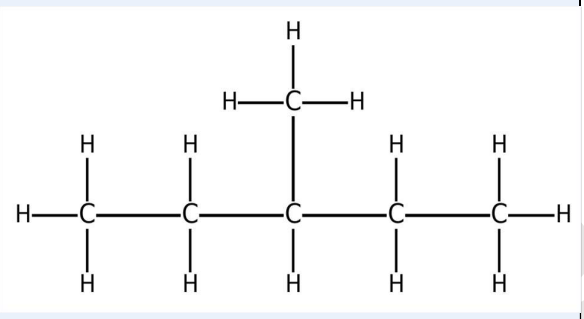
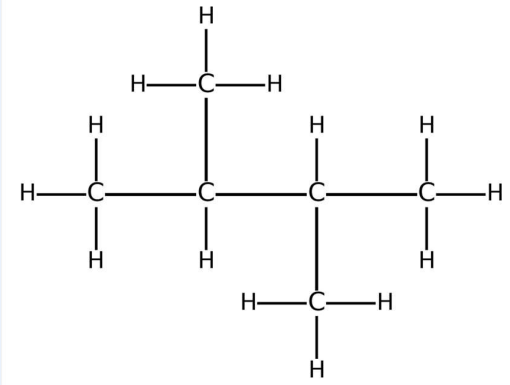
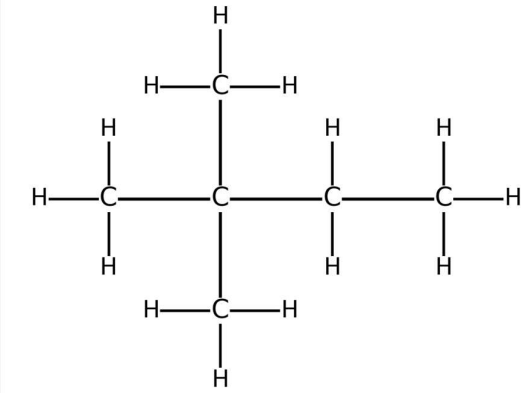
✓ **Alkanes (Paraffins)**

- © Saturated aliphatic hydrocarbons containing only C–C and C–H single (sigma) bonds.
- © They are saturated hydrocarbons.
- © General formula: CnH2n+2.
- © IUPAC suffix: '-ane'.

S.No	Common Name	IUPAC Name	Formula	Expanded Structure
1	Marsh Gas / Methyl Hydride	Methane	CH_4	
2	Dimethyl / Ethyl Hydride	Ethane	C_2H_6	
3	Propyl Hydride	Propane	C_3H_8	
4	n-Butane	Butane	C_4H_{10}	
4a	Isobutane (isomer of Butane)	2-Methylpropane	C_4H_{10}	

Conceptual Learning

S.No	Common Name	IUPAC Name	Formula	Expanded Structure
5	n-Pentane / Amyl Hydride	Pentane	C_5H_{12}	
5a	Isopentane (isomer of Pentane)	2-Methylbutane	C_5H_{12}	
5b	Neopentane (isomer of Pentane)	2,2-Dimethylpropane	C_5H_{12}	
6	n-Hexane	Hexane	C_6H_{14}	
6a	Isohexane (isomer of Hexane)	2-Methylpentane	C_6H_{14}	

S.No	Common Name	IUPAC Name	Formula	Expanded Structure
6b	Diethylmethylmethane (isomer of Hexane)	3-Methylpentane	C_6H_{14}	
6c	Di isopropyl (isomer of Hexane)	2,3-Dimethylbutane	C_6H_{14}	
6d	Neohexane (isomer of Hexane)	2,2-Dimethylbutane	C_6H_{14}	

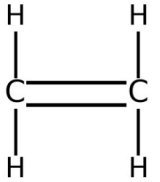
✓ **Alkenes(Olefins):**

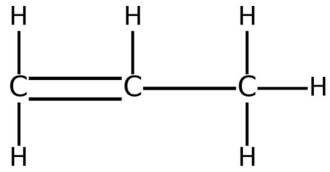
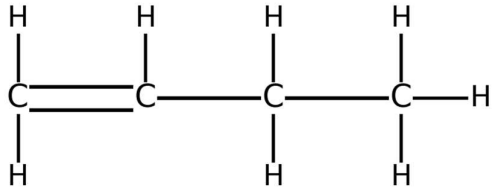
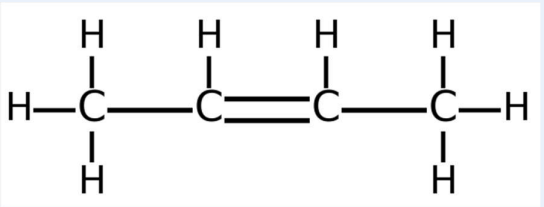
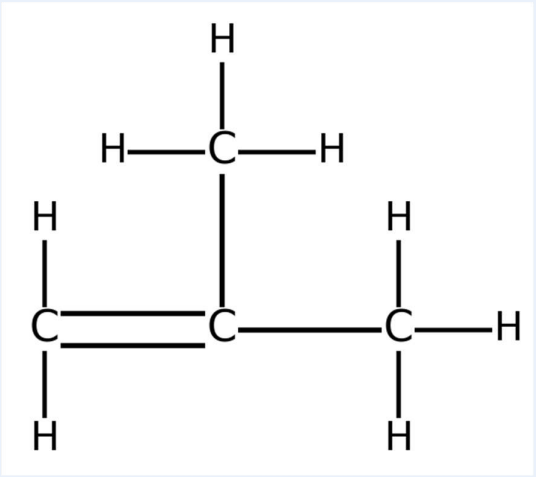
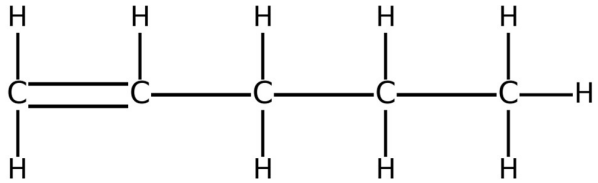
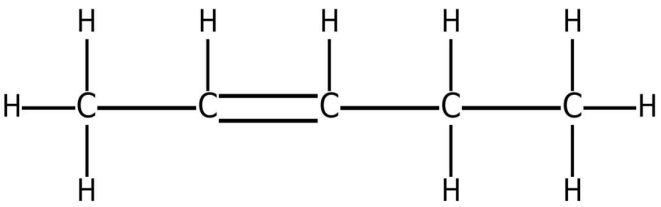
© Open-chain (acyclic) hydrocarbons with at least one carbon–carbon double bond.

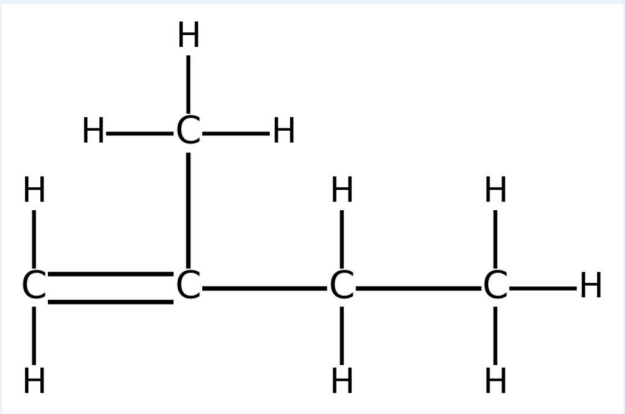
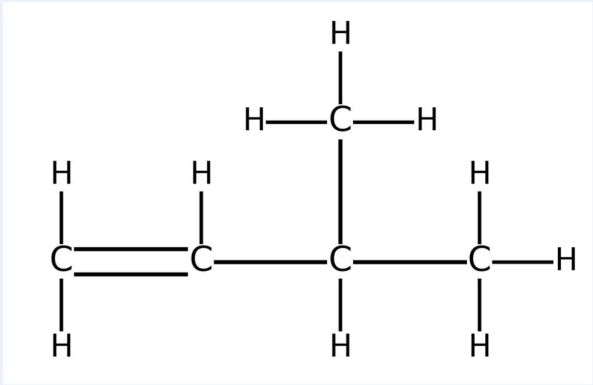
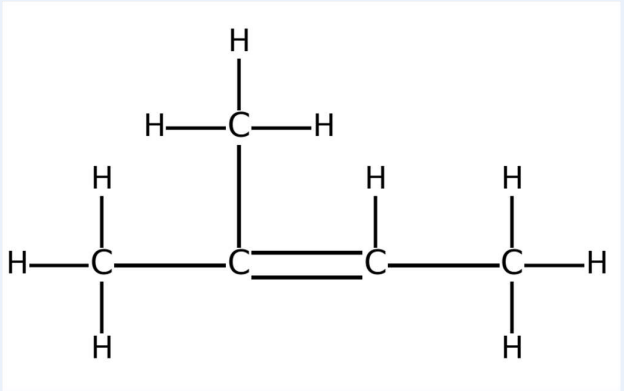
© They are unsaturated Hydrocarbons.

© General formula C_nH_{2n}.

© IUPAC suffix "ene"

S.No	Common Name	IUPAC Name	Formula	Expanded Structure (Large)
1	Ethylene	Ethene	C_2H_4	

S.No	Common Name	IUPAC Name	Formula	Expanded Structure (Large)
2	Propylene	Propene	C_3H_6	
3	α-Butylene	But-1-ene	C_4H_8	
3a	Cis or trans-Butylene (isomer of But-1-ene)	But-2-ene	C_4H_8	
3b	Isobutylene (isomer of But-1-ene)	2-Methylpropene	C_4H_8	
4	α-Amylene	Pent-1-ene	C_5H_{10}	
4a	β-Amylene (isomer of Pent-1-ene)	Pent-2-ene	C_5H_{10}	

S.No	Common Name	IUPAC Name	Formula	Expanded Structure (Large)
4b	unsym-Methylethylethylene (isomer of Pent-1-ene)	2-Methylbut-1-ene	C_5H_{10}	
4c	Isopropylethylene / Isopentene (isomer of Pent-1-ene)	3-Methylbut-1-ene	C_5H_{10}	
4d	Trimethylethylene (isomer of Pent-1-ene)	2-Methylbut-2-ene	C_5H_{10}	

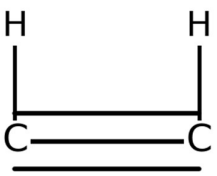
✓ **Alkynes:**

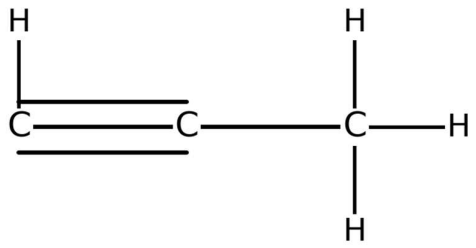
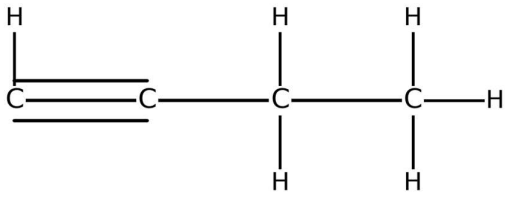
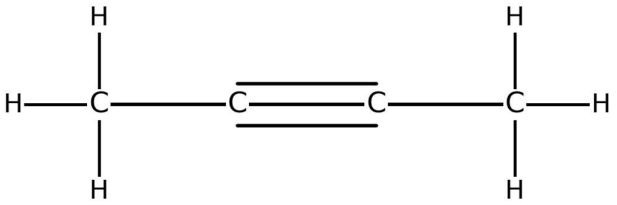
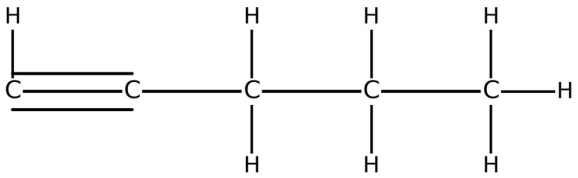
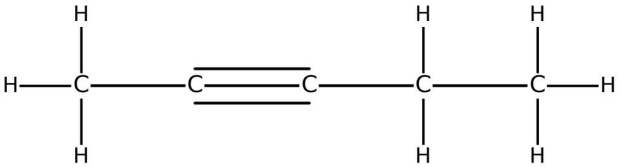
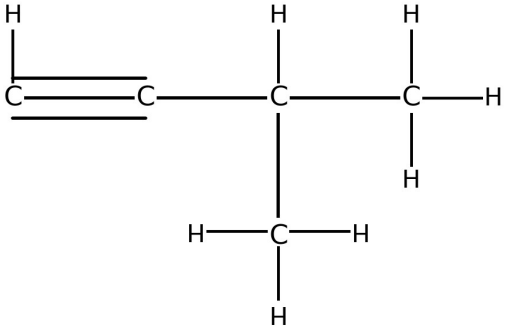
© Open-chain (acyclic) hydrocarbons with at least one carbon–carbon triple bond.

© They are unsaturated Hydrocarbons.

© General formula C_nH_{2n-2}.

© IUPAC suffix "yne"

S.No	Common Name	IUPAC Name	Formula	Expanded Structure (Large)
1	Acetylene	Ethyne	C_2H_2	

S.No	Common Name	IUPAC Name	Formula	Expanded Structure (Large)
2	Methylacetylene / Allylene	Propyne	C_3H_4	
3	Ethylacetylene	But-1-yne	C_4H_6	
3a	Dimethylacetylene / Crotonylene (isomer of But-1-yne)	But-2-yne	C_4H_6	
4	Propylacetylene	Pent-1-yne	C_5H_8	
4a	Methylethylacetylene (isomer of Pent-1-yne)	Pent-2-yne	C_5H_8	
4b	Isopropylacetylene (isomer of Pent-1-yne)	3-Methylbut-1-yne	C_5H_8	

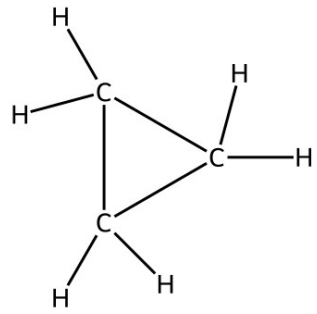
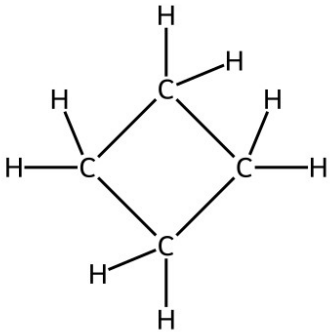
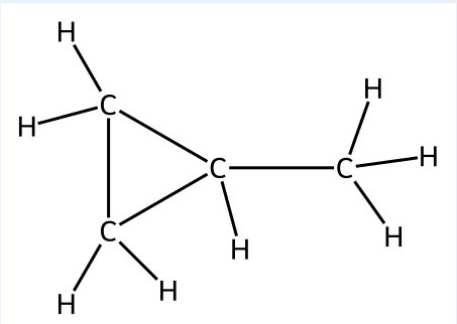
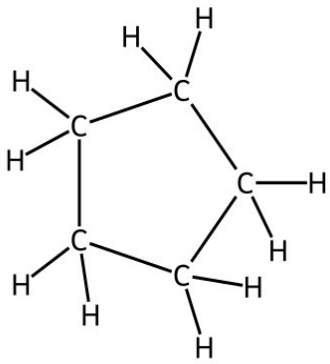
✓ **Cycloalkanes:**

© Cyclic compounds with single bond between carbon and carbon atom.

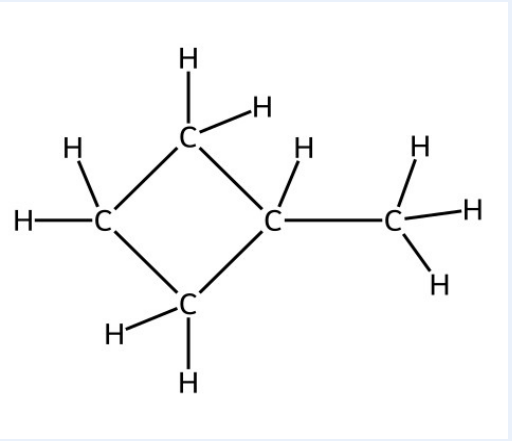
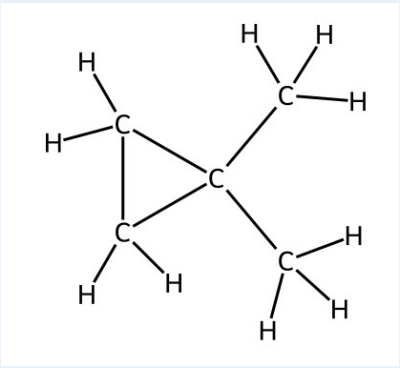
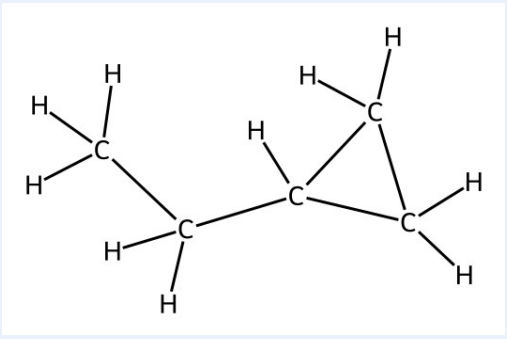
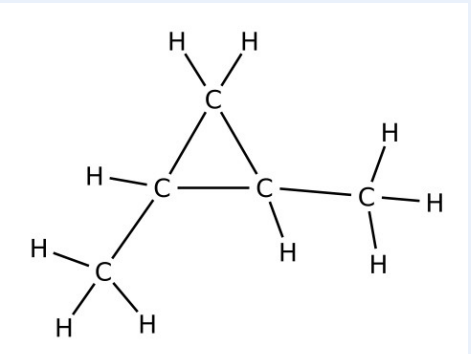
© They are saturated Hydrocarbons.

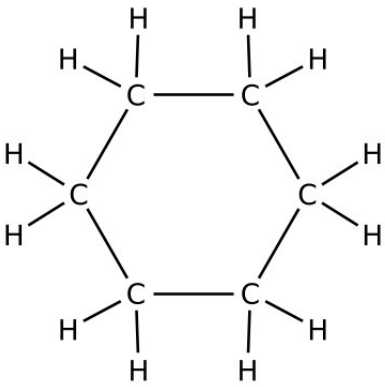
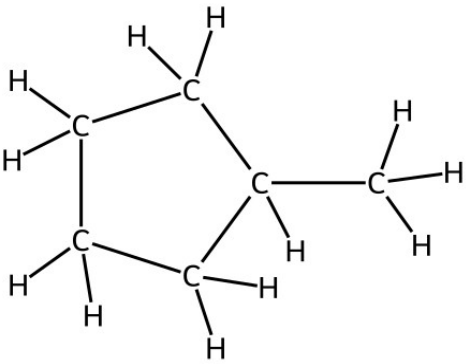
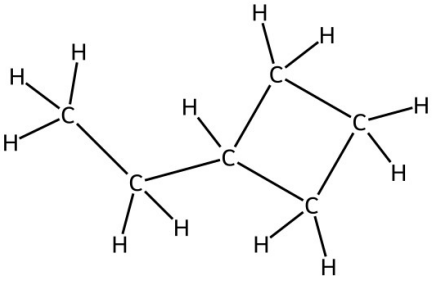
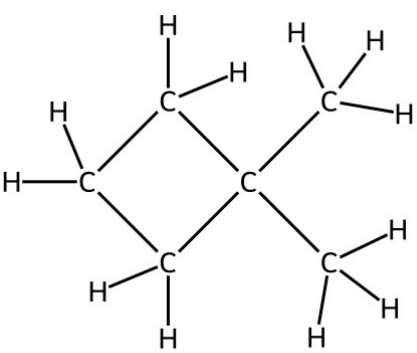
© General formula C_nH_{2n}.

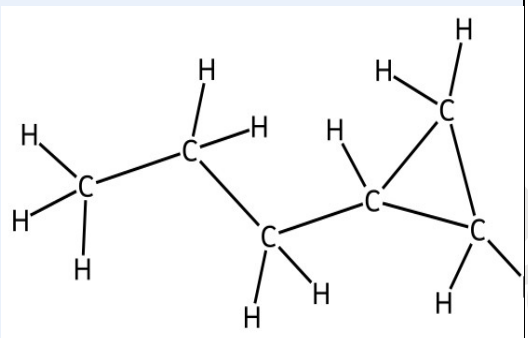
© IUPAC Prefix "Cyclo"

S.No	Common Name	IUPAC Name	Formula	Expanded Structure (Large)
1	Cyclopropane	Cyclopropane	C_3H_6	
2	Cyclobutane	Cyclobutane	C_4H_8	
2a	Methylcyclopropane (isomer of Cyclobutane)	Methylcyclopropane	C_4H_8	
3	Cyclopentane	Cyclopentane	C_5H_{10}	

Conceptual Learning

S.No	Common Name	IUPAC Name	Formula	Expanded Structure (Large)
3a	Methylcyclobutane (isomer of Cyclopentane)	Methylcyclobutane	C_5H_{10}	
3b	gem-Dimethylcyclopropane (isomer of Cyclopentane)	1,1-Dimethylcyclopropane	C_5H_{10}	
3c	Ethylcyclopropane (isomer of Cyclopentane)	Ethylcyclopropane	C_5H_{10}	
3d	1,2-Dimethylcyclopropane (isomer of Cyclopentane)	1,2-Dimethylcyclopropane	C_5H_{10}	

S.No	Common Name	IUPAC Name	Formula	Expanded Structure (Large)
4	Cyclohexane	Cyclohexane	C_6H_{12}	
4a	Methylcyclopentane (isomer of Cyclohexane)	Methylcyclopentane	C_6H_{12}	
4b	Ethylcyclobutane (isomer of Cyclohexane)	Ethylcyclobutane	C_6H_{12}	
4c	gem-Dimethylcyclobutane (isomer of Cyclohexane)	1,1-Dimethylcyclobutane	C_6H_{12}	

S.No	Common Name	IUPAC Name	Formula	Expanded Structure (Large)
4d	Propylcyclopropane (isomer of Cyclohexane)	Propylcyclopropane	C_6H_{12}	

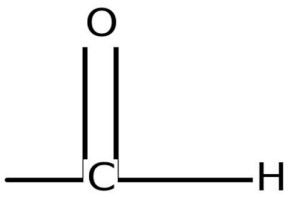
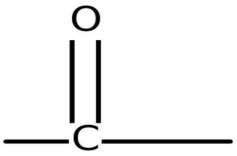
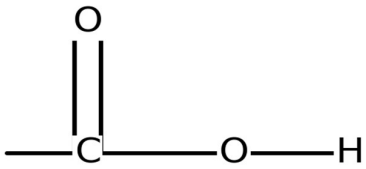
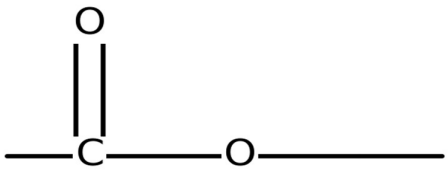
Note: Cycloalkanes are isomers of alkenes with the same number of carbon atoms (both are C_nH_{2n}, e.g. cyclopropane and propene are both C_3H_6). The C_6H_{12} rows above show only some of the possible cyclic isomers.

- ✓ **Alkyl Group**
  - © An alkyl group is simply an alkane that has lost one hydrogen, so it can attach to something else.
  - © Its name is formed by changing the alkane's '-ane' ending to '-yl'.
  - © **Examples:**
    - Methane (CH4) → Methyl group (–CH3)
    - Ethane (C2H6) → Ethyl group (–C2H5)
    - Propane (C3H8) → Propyl group (–C3H7)
  - © Alkyl Group is represented by "-**R"**
  - © General formula of an alkyl group: **C_nH_{2n+1}–**. The free line (–) shows that one valency of the carbon atom is free.
  - © So when R–O–H is written for an alcohol, replacing R with methyl (–CH3) gives CH3–OH, which is methanol — a real, complete molecule.
  - © For groups with two attachment points (ketones, ethers, esters), you will sometimes see **R and R'** (R-prime) used together — this just means the two alkyl groups attached can be the same or different.

❖ Functional Groups

- ✓ **Heteroatom:** carbon and hydrogen are the normal constituents of organic compounds. An atom other than carbon and hydrogen that replaces hydrogen in a carbon chain is called a **heteroatom** (hetero = different), e.g. halogens (Cl, Br, I), oxygen, nitrogen and sulphur.
- ✓ Heteroatoms, and groups containing them, give specific properties to the compound regardless of the length of the carbon chain – these are called **functional groups**. A saturated hydrocarbon is unreactive, but introducing a functional group makes it reactive.
- ✓ A functional group is a specific atom or group of atoms within a molecule that is responsible for that molecule's characteristic chemical reactions.
- ✓ The rest of the carbon chain mostly acts as a "backbone" of compound.
- ✓ The functional group decides how the molecule reacts.
- ✓ Compounds with the same functional group show similar chemical properties, regardless of chain length (this is the basis of a homologous series).
- ✓ For Example: The –OH group in ethanol (CH_3CH_2OH) is the functional group responsible for its characteristic reactions (e.g. oxidation to ethanal, esterification).
- ✓ The carbon–carbon double bond (**alkene group, C=C**) and triple bond (**alkyne group, C≡C**) are also treated as functional groups.

✓ Important Functional Groups are mentioned in the table below.

Functional Group (replaces H)	Class Name	General Formula	Expanded Structure
–OH	Alcohol	R–O–H	
–CHO	Aldehyde	R–CHO	
>C=O	Ketone	R–CO–R'	
–COOH	Carboxylic Acid	R–COOH	
–COOR	Ester	R–COO–R'	
–O–	Ether	R–O–R'	
–X (F/Cl/Br/I)	Halide (Haloalkane)	R–X	

✓ **Nomenclature of Hydrocarbons with Functional Groups:**

© Suffix and prefix of 11 functional groups **(Highest to Lowest priority)** attached with carbon compounds are shown in table below:

Priority	Group	Suffix	Prefix (if not suffix)
1	Carboxylic acid –COOH	-oic acid	carboxy-
2	Ester –COOR	-oate	alkoxycarbonyl-

Priority	Group	Suffix	Prefix (if not suffix)
3	Amide –CONH2	-amide	carbamoyl-
4	Nitrile –C≡N	-nitrile	cyano-
5	Aldehyde –CHO	-al	oxo-
6	Ketone >C=O	-one	oxo-
7	Alcohol –OH	-ol	hydroxy-
8	Amine –NH2	-amine	amino-
9	Ether –O–	(none)	alkoxy-
10	Halide –X	(none)	halo- (fluoro/chloro/bromo/iodo)
11	Nitro –NO2	(none)	nitro-

© **Rules for naming Hydrocarbons with functional groups:**

(1) Pick the Suffix (Principal Group)

⇒ Find the highest-priority group from the table above. That group becomes the suffix. Every other group becomes a prefix.

⇒ ***Example:*** *–COOH + –OH together → acid wins → suffix '-oic acid'; –OH becomes prefix 'hydroxy-'.*

(2) Number for the Group First

⇒ Numbering priority is now: (1) principal group, (2) double/triple bond, (3) substituents.

⇒ Example: Propanoic acid — the acid carbon is always C1, ahead of everything else.

(3) Rings as Parent Chain or Substituent

⇒ If the principal group sits directly on the ring, the ring is the parent. If the group is on a side chain, the chain is the parent and the ring becomes a prefix ('cyclohexyl-').

⇒ Example: Cyclohexanol (OH on ring) vs Cyclohexylmethanol (OH on the attached chain).

(4) Unsaturation + a Functional Group Together

⇒ The functional group keeps the suffix. Any C=C/C≡C is inserted as an infix ('-en-'/'-yn-') just before that suffix.

⇒ Example: But-3-en-1-ol (OH fixed at C1, double bond at C3); But-2-enoic acid (COOH fixed at C1).

(5) Assemble the Final Name

⇒ Order: alphabetised prefixes + parent + infix (if any) + suffix. No locant needed for a group fixed at C1 (acid, aldehyde, nitrile).

⇒ Example: Propanoic acid, not '1-propanoic acid'.

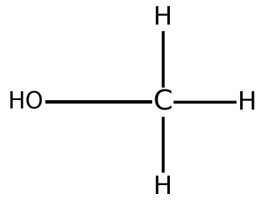
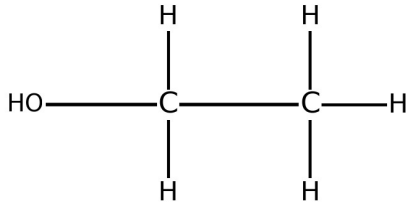
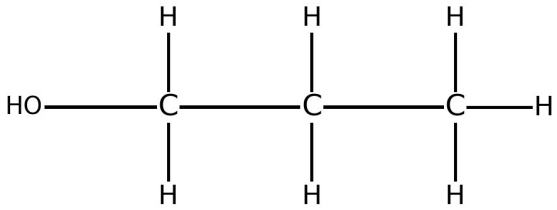
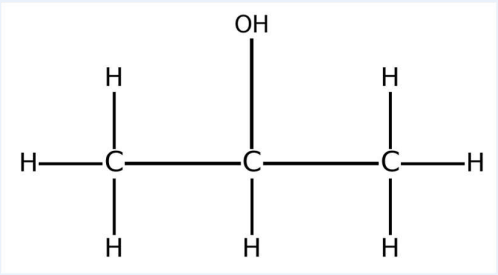
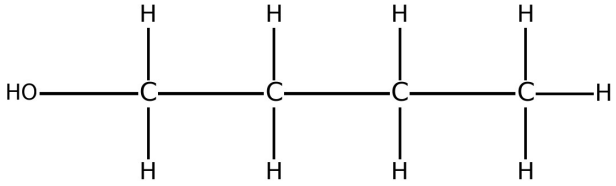
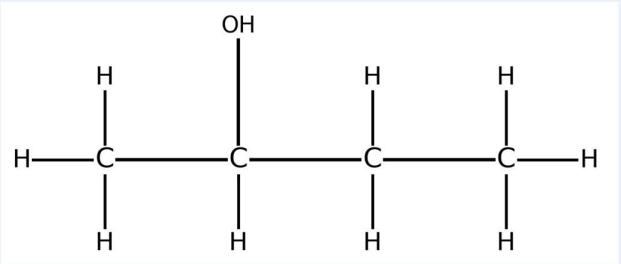
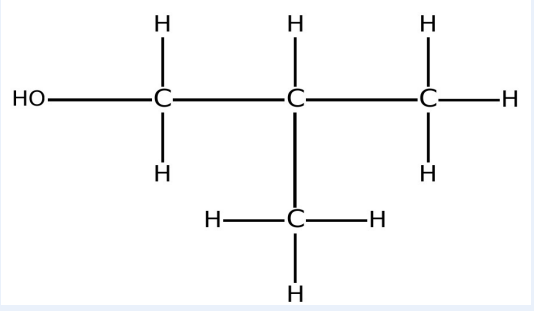
✓ Alcohols:

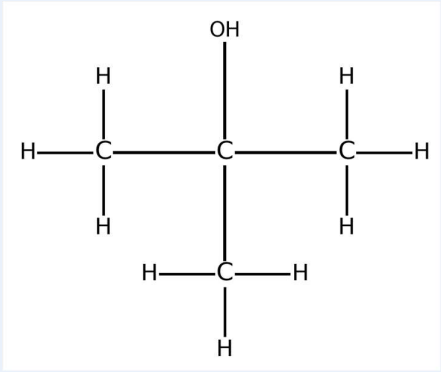
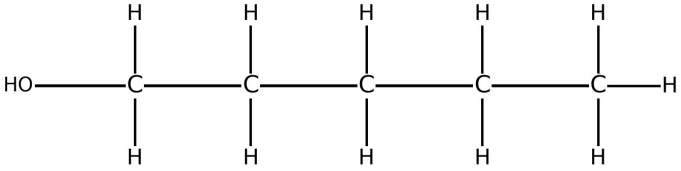
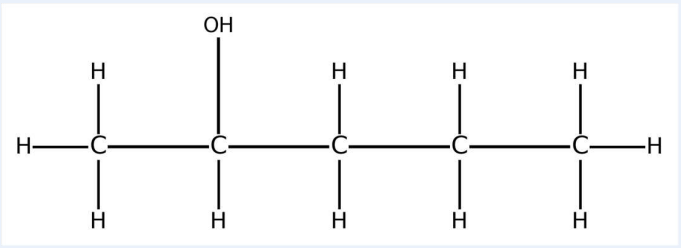
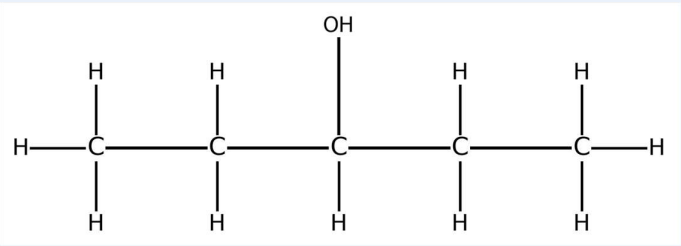
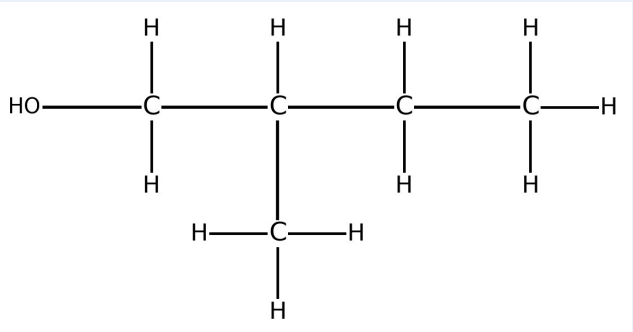
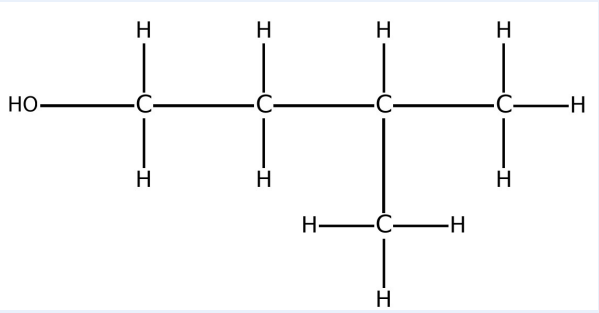
© An organic compound containing a hydroxyl group (–OH) attached to a carbon atom of an alkyl chain.

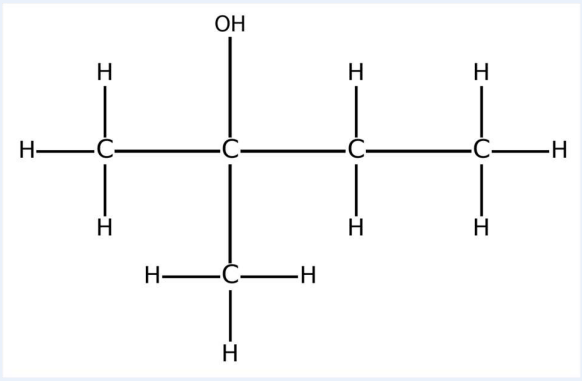
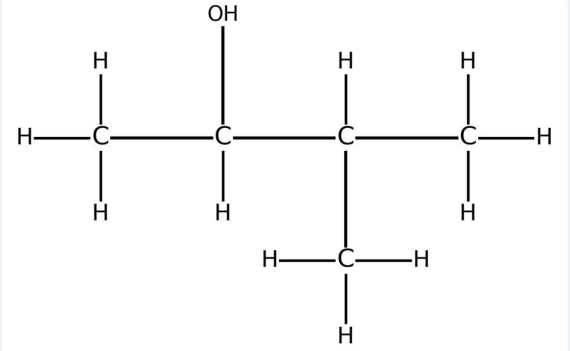
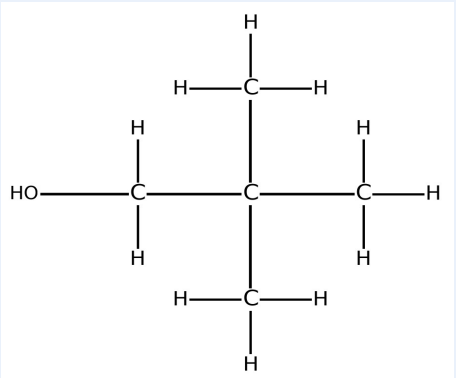
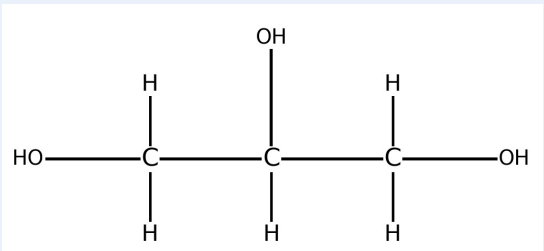
© General formula: $C_nH_{2n+1}OH$ (or R–OH, where R = alkyl group).

© IUPAC suffix: -ol (e.g., Methanol, Ethanol).

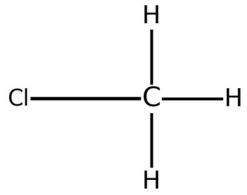
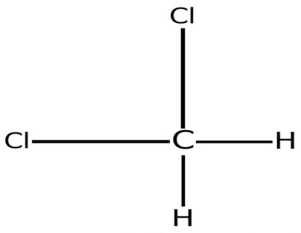
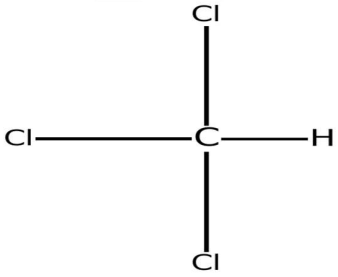
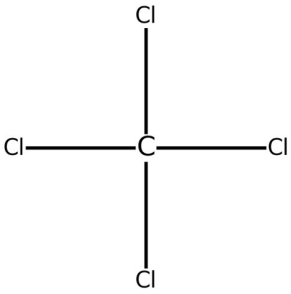
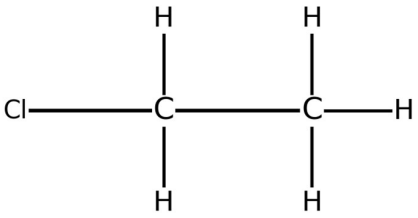
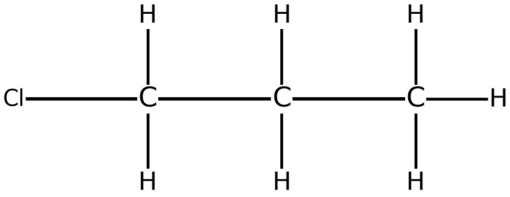
Conceptual Learning

S.No	Common Name	IUPAC Name	Formula	Expanded Structure
1	Wood Spirit / Methyl Alcohol	Methanol	CH_4O	
2	Grain Alcohol / Ethyl Alcohol	Ethanol	C_2H_6O	
3	n-Propyl Alcohol	Propan-1-ol	C_3H_8O	
3a	Isopropyl Alcohol	Propan-2-ol (isomer of Propan-1-ol)	C_3H_8O	
4	n-Butyl Alcohol	Butan-1-ol	$C_4H_{10}O$	
4a	sec-Butyl Alcohol	Butan-2-ol (isomer of Butan-1-ol)	$C_4H_{10}O$	
4b	Isobutyl Alcohol	2-Methylpropan-1-ol (isomer of Butan-1-ol)	$C_4H_{10}O$	

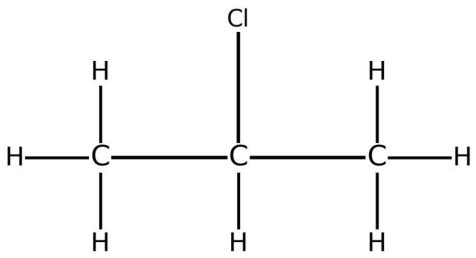
S.No	Common Name	IUPAC Name	Formula	Expanded Structure
4c	tert-Butyl Alcohol	2-Methylpropan-2-ol (isomer of Butan-1-ol)	$C_4H_{10}O$	
5	n-Pentyl Alcohol / Amyl Alcohol	Pentan-1-ol	$C_5H_{12}O$	
5a	sec-Pentyl Alcohol	Pentan-2-ol (isomer of Pentan-1-ol)	$C_5H_{12}O$	
5b	Diethylcarbinol	Pentan-3-ol (isomer of Pentan-1-ol)	$C_5H_{12}O$	
5c	Active Amyl Alcohol	2-Methylbutan-1-ol (isomer of Pentan-1-ol)	$C_5H_{12}O$	
5d	Isoamyl Alcohol	3-Methylbutan-1-ol (isomer of Pentan-1-ol)	$C_5H_{12}O$	

S.No	Common Name	IUPAC Name	Formula	Expanded Structure
5e	tert-Amyl Alcohol	2-Methylbutan-2-ol (isomer of Pentan-1-ol)	$C_5H_{12}O$	
5f	sec-Isoamyl alcohol	3-Methylbutan-2-ol (isomer of Pentan-1-ol)	$C_5H_{12}O$	
5g	Neopentyl Alcohol	2,2-Dimethylpropan-1-ol (isomer of Pentan-1-ol)	$C_5H_{12}O$	
One Important Alcohol				
6	Glycerol / Glycerin	Propane-1,2,3-triol (3 –OH, trihydric)	$C_3H_8O_3$	

- ✓ **Haloalkanes (Halides):**
  - © A haloalkane (alkyl halide) is formed when one or more hydrogen atoms of an alkane are replaced by a halogen atom like Fluorine (F), Chlorine (Cl), Bromine (Br), or Iodine (I).
  - © General formula: $C_nH_{2n+1}X$ for monohaloalkanes (one halogen atom), often written as R–X, where R is an alkyl group and X is the halogen. Compounds with more than one halogen (e.g. CH_2Cl_2, $CHCl_3$, CCl_4) do not fit this formula.
  - © For IUPAC naming, Halogens are always named as a prefix (fluoro-, chloro-, bromo-, iodo-) with locant. They can never be the principal characteristic group of a name.
  - © Example: CH3Cl is Chloromethane; the '–Cl' is cited as a prefix, not a suffix.

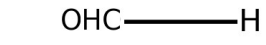
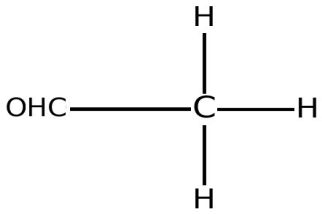
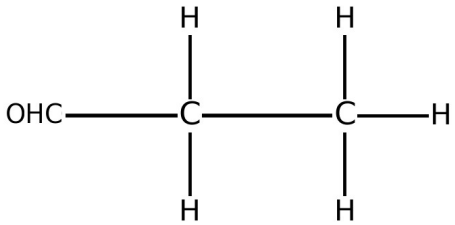
S.No	Common Name	IUPAC Name	Formula	Expanded Structure
1	Methyl Chloride	Chloromethane (1 chlorine)	CH_3Cl	
2	Methylene Chloride	Dichloromethane (2 chlorines)	CH_2Cl_2	
3	Chloroform	Trichloromethane (3 chlorines)	$CHCl_3$	
4	Carbon Tetrachloride	Tetrachloromethane (4 chlorines)	CCl_4	
5	Ethyl Chloride	Chloroethane	C_2H_5Cl	
6	n-Propyl Chloride	1-Chloropropane	C_3H_7Cl	

Conceptual Learning

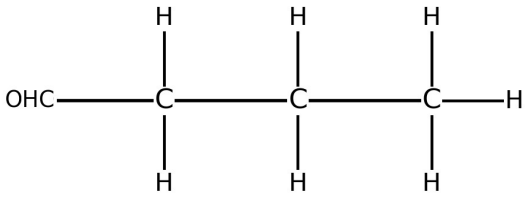
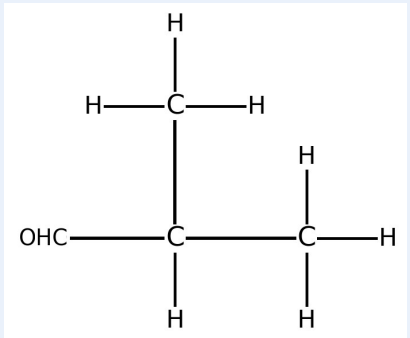
S.No	Common Name	IUPAC Name	Formula	Expanded Structure
6a	Isopropyl Chloride	2-Chloropropane (isomer of 1-Chloropropane)	C_3H_7Cl	

✓ Aldehydes (–CHO)

- © An aldehyde has the carbonyl group (C=O) attached to at least one hydrogen atom, always at the end of the carbon chain.
- © General formula: $C_nH_{2n}O$. IUPAC suffix: '-al'.
- © Because the carbonyl carbon is fixed at C1(First carbon of chain), no locant is ever written for it (e.g. 'Propanal', never '1-propanal').
- © Physical properties: lower aldehydes (methanal, ethanal) are gases/volatile liquids with a sharp smell; solubility in water decreases as the chain lengthens.
- © Key reactions: easily OXIDISED to a carboxylic acid, reduced to a primary alcohol; undergoes addition reactions across the C=O bond.
- © Common examples: Formaldehyde (preservative, resins), Acetaldehyde (industrial intermediate).
- © **Uses of methanal:** used for making plastics (e.g., for electric plugs and sockets); a solution of methanal in water, called **formalin**, is used to **preserve biological specimens**.

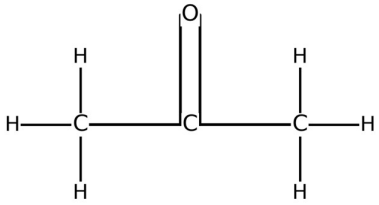
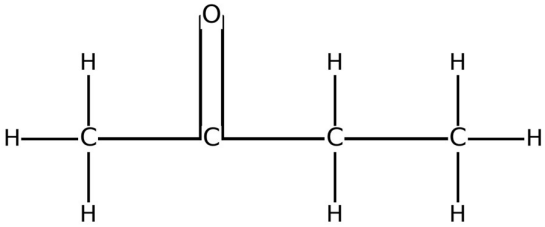
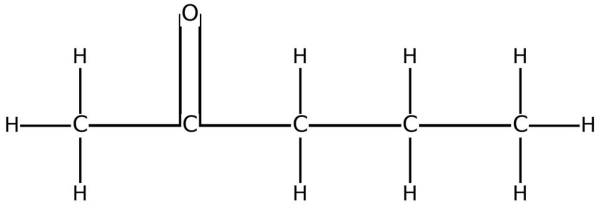
S.No	Common Name	IUPAC Name	Formula	Expanded Structure
1	Formaldehyde	Methanal	CH_2O	
2	Acetaldehyde	Ethanal	C_2H_4O	
3	Propionaldehyde	Propanal	C_3H_6O	

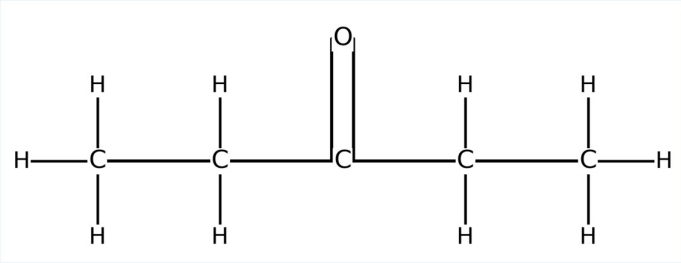
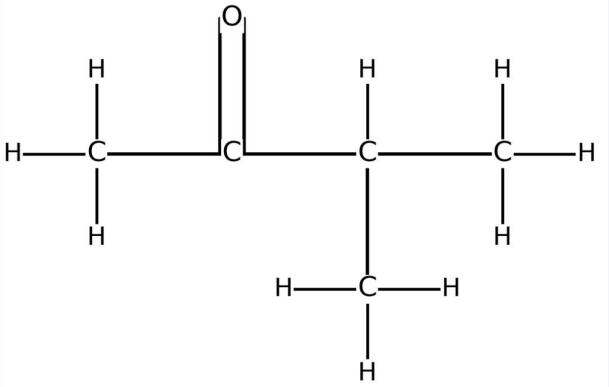
Conceptual Learning

S.No	Common Name	IUPAC Name	Formula	Expanded Structure
4	n-Butyraldehyde	Butanal	C_4H_8O	
4a	Isobutyraldehyde	2-Methylpropanal (isomer of Butanal)	C_4H_8O	

✓ Ketones (>C=O)

- © Ketone has carbonyl group (C=O) attached to two carbon atoms, so it can never sit at the end of the chain.
- © General formula: $C_nH_{2n}O$ (same as aldehyde).
- © IUPAC suffix: '-one'.
- © Ketones need minimum of 3 carbons to exist as the carbonyl carbon must be bonded to two other carbons.
- © Key reactions: resistant to mild oxidation (unlike aldehydes, ketones are not easily oxidised further), reduced to a secondary alcohol; undergoes addition reactions across the C=O bond.
- © Common examples: Acetone (nail polish remover, solvent), Methyl ethyl ketone / MEK (industrial solvent).

S.No	Common Name	IUPAC Name	Formula	Expanded Structure
1	Acetone / Dimethyl Ketone	Propanone	C_3H_6O	
2	Methyl Ethyl Ketone (MEK)	Butanone	C_4H_8O	
3	Methyl Propyl Ketone	Pentan-2-one	$C_5H_{10}O$	

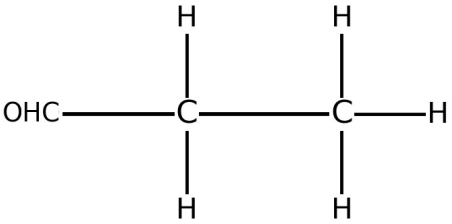
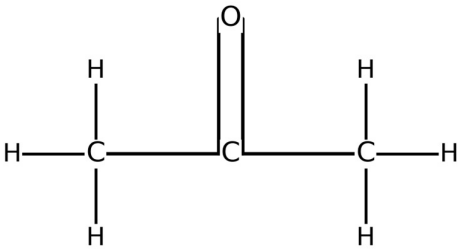
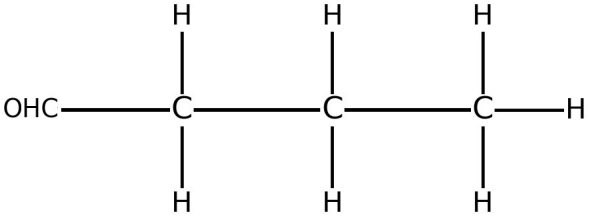
S.No	Common Name	IUPAC Name	Formula	Expanded Structure
3a	Diethyl Ketone	Pentan-3-one (isomer of Pentan-2-one)	$C_5H_{10}O$	
3b	Methyl Isopropyl Ketone	3-Methylbutan-2-one (isomer of Pentan-2-one)	$C_5H_{10}O$	

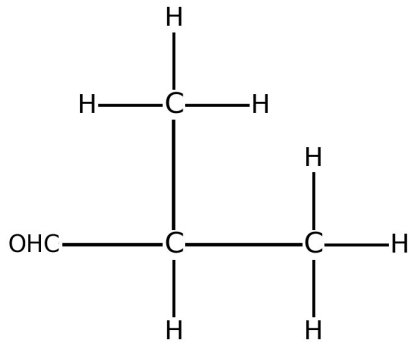
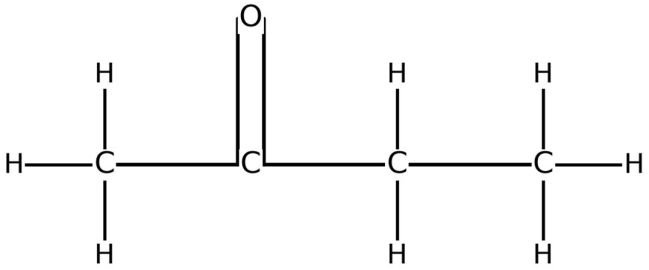
Note: Aldehydes and Ketones Are Isomers

© Aldehydes and ketones share the exact same general formula, CnH2nO.

© For the same number of carbons (from C3 onward), an aldehyde and a ketone can have the identical molecular formula while being completely different classes of compound.

© Such isomerism is called functional (group) isomerism, and the compounds are called functional isomers.

S.No	Common Name	IUPAC Name	Formula	Expanded Structure
1	Propion aldehyde	Propanal (aldehyde)	C_3H_6O	
1a	Acetone / Dimethyl Ketone	Propanone (ketone — functional isomer of Propanal)	C_3H_6O	
2	n-Butyraldehyde	Butanal (aldehyde)	C_4H_8O	

S.No	Common Name	IUPAC Name	Formula	Expanded Structure
2a	Isobutyraldehyde	2-Methylpropanal (aldehyde — chain isomer of Butanal)	C_4H_8O	
2b	Methyl Ethyl Ketone (MEK)	Butanone (ketone — functional isomer of Butanal)	C_4H_8O	

✓ Carboxylic Acids (–COOH)

© A carboxylic acid has the –COOH group (a carbonyl C=O and a hydroxyl –OH on the same carbon), always at C1 of the chain.

© General formula: $C_nH_{2n}O_2$. IUPAC suffix: '-oic acid'.

© Carboxylic acids are the HIGHEST priority functional group. If a COOH is present, it always becomes the suffix, and every other group becomes a prefix.

© Carboxylic acids are also called **organic acids** or **alkanoic acids**. Their general formula is **R–COOH**; only for the simplest acid, formic acid (HCOOH), R is a hydrogen atom.

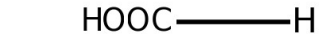
© Physical properties:

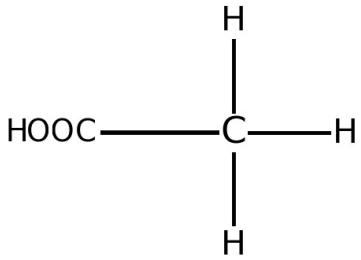
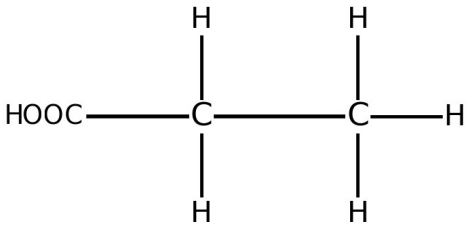
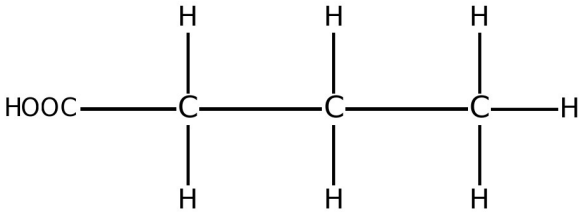
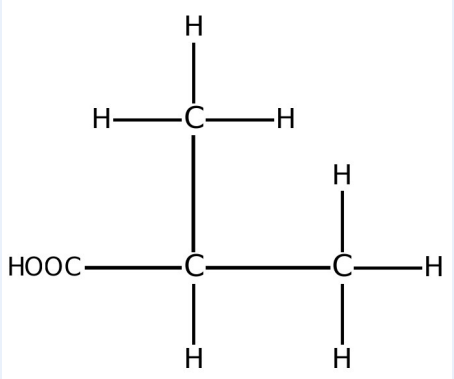
- Form strong hydrogen bonds
- High boiling point
- Lower acids are pungent, sour-smelling liquids miscible with water.

© Key reactions:

- It is Acidic and turns blue litmus red.
- It reacts with bases to form salts, so undergo neutralization reaction.
- It reacts with carbonates/bicarbonates releasing CO_2 gas (brisk effervescence).
- It forms esters with alcohols and the reaction is called Esterification.
- It can be reduced to a primary alcohol (using a strong reducing agent such as $LiAlH_4$).

© Common examples: Formic acid (ant and nettle stings), Acetic acid (vinegar), Citric acid and other natural acids.

S.No	Common Name	IUPAC Name	Formula	Expanded Structure
1	Formic Acid	Methanoic Acid	CH_2O_2	

S.No	Common Name	IUPAC Name	Formula	Expanded Structure
2	Acetic Acid	Ethanoic Acid	$C_2H_4O_2$	
3	Propionic Acid	Propanoic Acid	$C_3H_6O_2$	
4	n-Butyric Acid	Butanoic Acid	$C_4H_8O_2$	
4a	Isobutyric Acid	2-Methylpropanoic Acid (isomer of Butanoic Acid)	$C_4H_8O_2$	

✓ Esters (–COOR)

© An ester is formed when a carboxylic acid reacts with an alcohol, releasing water — this reaction is called esterification.

© General formula: R–COO–R', where R and R' are alkyl groups (from the acid and alcohol respectively). IUPAC suffix: '-oate'. For esters made from saturated acids and alcohols: $C_nH_{2n}O_2$ (same as carboxylic acids).

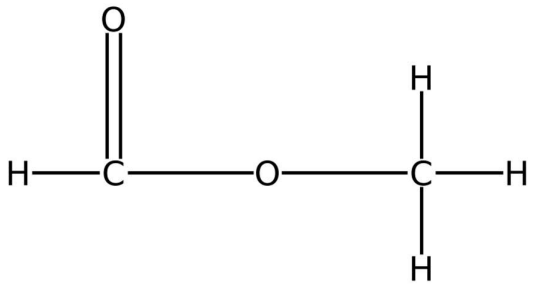
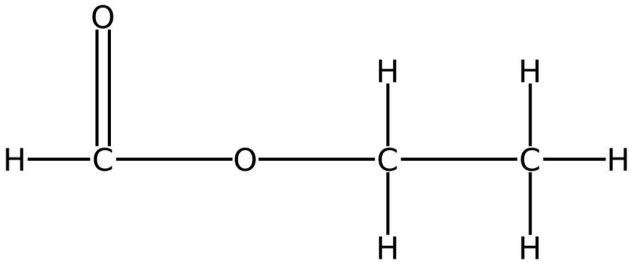
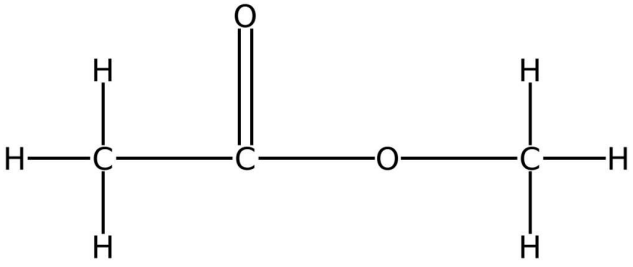
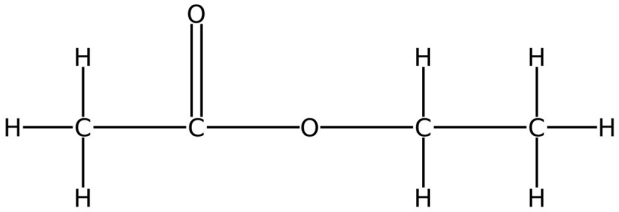
© Named as 'alkyl alkanoate' — the alcohol part is named first (as 'alkyl'), followed by the acid part (as 'alkanoate').

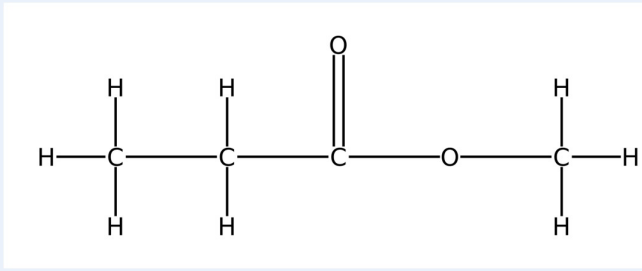
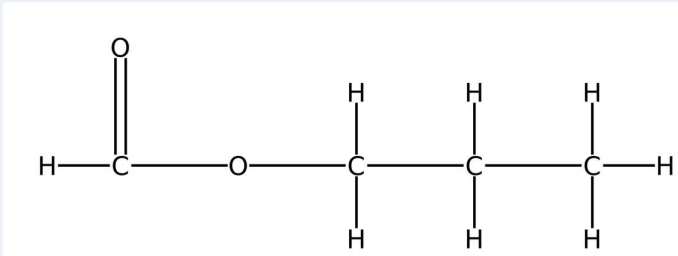
© Example: $CH_3COOC_2H_5$ is ethyl ethanoate — "ethyl" from ethanol (alcohol part) and "ethanoate" from ethanoic acid (acid part).

© Physical properties: esters are usually pleasant, fruity or flower-like smelling liquids; they are only slightly soluble in water, but dissolve well in organic solvents.

© Key reactions: hydrolysis (with water/acid or base) breaks the ester back into its acid and alcohol; with a base like NaOH, this reverse reaction is called saponification (the basis of soap-making).

- © Common examples and uses: esters give many fruits and flowers their characteristic smell, and are used to make artificial flavourings, perfumes, and fragrances.
- © Esters are **volatile** liquids with a sweet, fruity smell. They are used in artificial perfumes and as flavouring agents in **ice-cream, sweets and cold drinks**.
- © **Finding the parent acid and alcohol of an ester:** the part containing CO comes from the acid and the other part from the alcohol. E.g. $C_3H_7COOC_2H_5$ (ethyl butanoate) is made from **butanoic acid** (C_3H_7COOH) and **ethanol** (C_2H_5OH).

S.No	Common Name	IUPAC Name	Formula	Expanded Structure
1	Methyl Formate	Methyl methanoate	$C_2H_4O_2$	
2	Ethyl Formate	Ethyl methanoate	$C_3H_6O_2$	
2a	Methyl Acetate	Methyl ethanoate (isomer of Ethyl methanoate)	$C_3H_6O_2$	
3	Ethyl Acetate	Ethyl ethanoate	$C_4H_8O_2$	

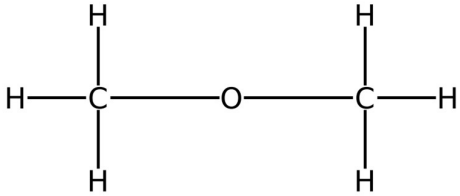
S.No	Common Name	IUPAC Name	Formula	Expanded Structure
3a	Methyl Propionate	Methyl propanoate (isomer of Ethyl ethanoate)	$C_4H_8O_2$	
3b	n-Propyl Formate	Propyl methanoate (isomer of Ethyl ethanoate)	$C_4H_8O_2$	

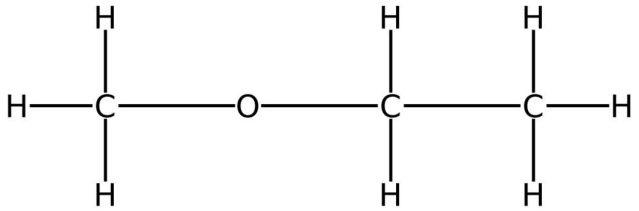
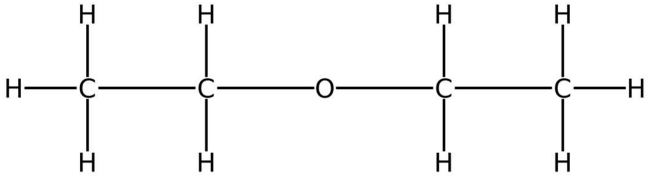
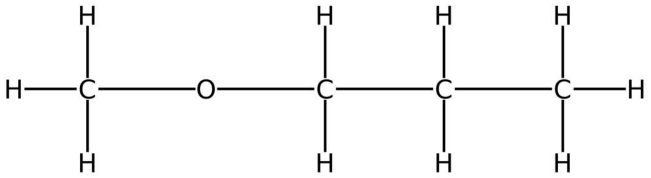
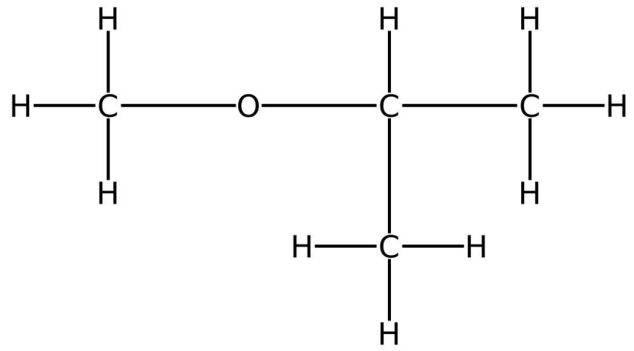
Note: Carboxylic Acids and Esters Are Isomers

- © Carboxylic acids and esters share the same general formula, $C_nH_{2n}O_2$.
- © For the same number of carbons (from C2 onward), an acid and an ester can have the identical molecular formula, e.g. Ethanoic acid (CH_3COOH) and Methyl methanoate ($HCOOCH_3$) are both $C_2H_4O_2$.
- © Such compounds are functional isomers.

✓ Ethers (–O–)

- © An ether has an oxygen atom bonded to two alkyl groups, written as R–O–R' (R and R' can be the same or different).
- © General formula: $C_nH_{2n+2}O$ (same as alcohols). Ethers have no IUPAC suffix.
- © IUPAC naming: named as 'alkoxyalkane' — the smaller alkyl group with the oxygen becomes the 'alkoxy' prefix (methoxy-, ethoxy-), and the larger group is the parent alkane (e.g. $CH_3–O–C_2H_5$ is Methoxyethane).
- © Common names: name both alkyl groups in alphabetical order, then add 'ether' (e.g. Ethyl methyl ether, Diethyl ether).
- © Physical properties: lower ethers are volatile, highly flammable, sweet-smelling gases/liquids; they have much lower boiling points than isomeric alcohols because ether molecules cannot form hydrogen bonds with each other; only slightly soluble in water.
- © Key reactions: fairly unreactive (so they are good solvents); burn in air to give CO_2 and H_2O; broken (cleaved) by strong acids such as HI.
- © Common examples: Diethyl ether (laboratory solvent, earlier used as an anaesthetic), Dimethyl ether (aerosol propellant, fuel).

S.No	Common Name	IUPAC Name	Formula	Expanded Structure
1	Dimethyl Ether	Methoxymethane	C_2H_6O	

S.No	Common Name	IUPAC Name	Formula	Expanded Structure
2	Ethyl Methyl Ether	Methoxyethane	C_3H_8O	
3	Diethyl Ether	Ethoxyethane	$C_4H_{10}O$	
3a	Methyl n-Propyl Ether	1-Methoxypropane (isomer of Ethoxyethane)	$C_4H_{10}O$	
3b	Methyl Isopropyl Ether	2-Methoxypropane (isomer of Ethoxyethane)	$C_4H_{10}O$	

Note: Alcohols and Ethers Are Isomers

- © Alcohols and ethers share the same general formula, $C_nH_{2n+2}O$.
- © For the same number of carbons (from C2 onward), an alcohol and an ether can have the identical molecular formula, e.g. Ethanol (C_2H_5OH) and Methoxymethane (CH_3OCH_3) are both C_2H_6O.
- © Such compounds are functional isomers.

❖ Physical Properties of Carbon Compounds

- © Most carbon compounds are **poor conductors of electricity** – the bonding in these compounds does not give rise to any ions (no charged particles are formed).
- © They have **low melting and boiling points** compared with ionic compounds. This shows that the forces of attraction between their molecules are not very strong.

- © Covalently bonded molecules have **strong bonds within the molecule**, but the **intermolecular forces are weak**. This is the reason for their low melting and boiling points.
- © Melting and boiling points of some carbon compounds (NCERT Table 4.1):

Compound	Melting point (K)	Boiling point (K)	State at room temperature
Acetic acid (CH_3COOH)	290 (17 °C)	391 (118 °C)	Liquid (freezes in winter)
Chloroform ($CHCl_3$)	209 (–64 °C)	334 (61 °C)	Liquid
Ethanol (CH_3CH_2OH)	156 (–117 °C)	351 (78 °C)	Liquid
Methane (CH_4)	90 (–183 °C)	111 (–162 °C)	Gas

- © **Physical state:** carbon compounds may be gases (methane, ethene), liquids (ethanol, chloroform, acetic acid) or solids (naphthalene, camphor, sugar, wax) at room temperature.
- © **Solubility:** most carbon compounds are insoluble or only slightly soluble in water but dissolve in organic solvents. Lower alcohols and acids are exceptions – ethanol is soluble in water in all proportions.
- © **Gradation in a homologous series:** as molecular mass increases, melting point, boiling point and solubility in a particular solvent change gradually, while the chemical properties (decided by the functional group) remain similar.
- © Most carbon compounds are **combustible** – they burn in air releasing heat and light. That is why carbon and its compounds are our major sources of fuels.

❖ Chemical Properties of Carbon Compounds

- © Since most of the fuels we use are either carbon or its compounds, we first study combustion. The four important chemical reactions of carbon compounds are – **Combustion, Oxidation, Addition and Substitution**.
- © **Combustion** takes place in all hydrocarbons (saturated and unsaturated); **substitution** reactions are given only by saturated hydrocarbons (alkanes); **addition** reactions are given only by unsaturated hydrocarbons (alkenes and alkynes).

(i) Combustion

- © Combustion is a chemical reaction in which a substance burns in oxygen to release heat and light. It is an oxidation reaction.
- © Carbon, in all its allotropic forms, burns in oxygen to give carbon dioxide along with the release of heat and light.
- © Most carbon compounds also release a large amount of heat and light on burning.
- © Alkanes burn in air producing a lot of heat, so they are **excellent fuels**. Natural gas (methane) is used as a fuel in homes, industry and vehicles (as CNG); **LPG** (mainly butane) is used as cooking gas.
- © **Examples of Combustion:**
  - Combustion of Carbon:
    $C + O_2 \rightarrow CO_2$ + heat and light
  - Combustion of Methane:
    $CH_4 + 2O_2 \rightarrow CO_2 + 2H_2O$ + heat and light
  - Combustion of Ethanol:
    $CH_3CH_2OH + 3O_2 \rightarrow 2CO_2 + 3H_2O$ + heat and light

✓ **Activity 4.3 – Burning of Carbon Compounds (teacher's assistance needed)**

(i) Procedure

- Take some carbon compounds (naphthalene, camphor, alcohol) one by one on a spatula and burn them.
- Observe the nature of the flame and note whether smoke is produced.
- Place a metal plate above the flame and check for any deposit.

(ii) Observation

- Naphthalene and camphor burn with a yellow, sooty flame giving lots of black smoke; a black (soot) deposit forms on the metal plate.
- Alcohol burns with a clean blue flame; there is no deposit on the plate.

(iii) Conclusion

- Compounds rich in carbon / unsaturated compounds burn with a sooty flame (incomplete combustion); saturated compounds such as alcohol burn with a clean flame.

✓ **Activity 4.4 – Flames of a Bunsen Burner**

(i) Procedure

- Light a Bunsen burner and adjust the air hole at its base to get different types of flames.
- Note when a yellow sooty flame and when a blue flame is obtained.

(ii) Observation

- Air hole closed (less air): yellow, sooty flame.
- Air hole open (enough air): clean blue flame.

(iii) Conclusion

- Limited supply of air causes incomplete combustion (sooty flame); enough air gives complete combustion (blue flame).

✓ **Clean Flame and Sooty Flame**

© **Saturated hydrocarbons** generally give a **clean flame**, while **unsaturated carbon compounds** give a **yellow flame with lots of black smoke**. This causes the sooty deposit on the metal plate in Activity 4.3.

© Limiting the supply of air results in **incomplete combustion** of even saturated hydrocarbons, giving a sooty flame.

© **Reason:** saturated hydrocarbons have a comparatively **low percentage of carbon**, which gets oxidised completely in air. Unsaturated hydrocarbons have a **higher percentage of carbon**, and air contains only about **21% oxygen**, which is not enough for their complete combustion.

© If unsaturated hydrocarbons are burnt in **pure oxygen**, they burn completely with a **blue flame** and no smoke.

© The gas/kerosene stove used at home has **inlets for air**, so that a sufficiently oxygen-rich mixture is burnt to give a **clean blue flame**.

© If the bottoms of cooking vessels are getting blackened, it means the air holes are blocked and fuel is getting wasted (the fuel is not burning completely).

© Fuels such as coal and petroleum contain some nitrogen and sulphur. Their combustion forms **oxides of sulphur and nitrogen**, which are major pollutants in the environment (they cause acid rain).

✓ **Disadvantages of Incomplete Combustion**

© Unburnt carbon forms **soot**, which pollutes the atmosphere.

© Soot **blackens cooking utensils** and **blocks chimneys** in factories.

© It produces **carbon monoxide (CO)**, an extremely poisonous gas.

© It produces **less heat** than complete combustion, so fuel is wasted.

✓ **Burning With and Without a Flame**

© Candle wax and LPG burn with a flame, but coal or charcoal in an 'angithi' sometimes just glows red and gives out heat **without a flame**.

© This is because **a flame is produced only when gaseous substances burn**. When wood or charcoal is ignited, the volatile substances present vaporise and burn with a flame in the beginning.

© A **luminous flame** is seen when the atoms of the gaseous substance are heated and start to glow. The colour produced by each element is characteristic of that element (e.g., a copper wire heated in a gas flame gives a bluish-green flame).

- © Incomplete combustion gives **soot**, which is carbon. The **yellow colour of a candle flame** is due to unburnt carbon (soot) particles that get heated and glow.

✓ **Types of Flames**

- © **Blue (non-luminous) flame:** when the supply of oxygen (air) is sufficient, fuels burn **completely**, producing a blue flame that gives little light. E.g. LPG in a gas stove, which has holes (inlets) for air to mix with the gas.
- © **Yellow (luminous) flame:** when the supply of oxygen is insufficient, fuels burn **incompletely**, producing a yellow, light-giving flame. The yellow colour is due to the glow of hot, **unburnt carbon particles**.
- © **Candle flame:** the wax melts, rises up the wick and turns into vapour. There is no arrangement for proper mixing of air, so the wax vapour burns incompletely; the unburnt carbon particles glow (yellow, luminous flame) and then leave as **soot and smoke**.
- © All **gaseous fuels** burn with a flame; solid and liquid fuels burn with a flame only if they **vaporise on heating** (e.g., wax, kerosene). Fuels that do not vaporise (coal, charcoal) burn without a flame.

✓ **Formation of Coal and Petroleum**

- © A **fuel** is a material that has energy stored in it; when burnt, it releases energy mainly as heat (and some light), used for cooking, heating, running vehicles, factories and power stations.
- © Fuels such as **coal, coke and charcoal** contain free carbon, while **kerosene, petrol, LPG and natural gas** are carbon compounds. Diamond and graphite are also free carbon but are not burnt as fuels.
- © Most fuels we use today are obtained from **coal, petroleum and natural gas** (fossil fuels).
- © Coal and petroleum have been formed from **biomass** that was subjected to various biological and geological processes.
- © **Coal** is the remains of trees, ferns and other plants that lived millions of years ago. They were crushed into the earth (perhaps by earthquakes or volcanic eruptions), pressed down by layers of earth and rock, and slowly decayed into coal.
- © **Oil and gas** are the remains of millions of tiny plants and animals that lived in the sea. Their dead bodies sank to the sea bed, were covered by silt, and bacteria turned them into oil and gas under high pressure. The silt was compressed into rock; oil and gas seeped into the porous rock and got trapped like water in a sponge.
- © They are called **fossil fuels** because they are formed from the remains (fossils) of living organisms buried millions of years ago. They are non-renewable.
- © **Coal** is a complex mixture of compounds of carbon, hydrogen and oxygen, with some free carbon and small amounts of nitrogen and sulphur compounds. It is obtained from **coal mines** deep under the earth. It was formed from buried forests under **high temperature and pressure** in the **absence of air**.
- © **Petroleum** means **rock oil** (petra = rock, oleum = oil) because it is found trapped in rocks under the earth's crust. It is a **dark-coloured, viscous, foul-smelling** crude oil – a mixture of solid, liquid and gaseous hydrocarbons mixed with water, salt and earth particles.
- © The petroleum formed got trapped between layers of **impervious (non-porous) rocks**, forming an **oil trap**; **natural gas** lies above the petroleum. Fuels like **petrol, kerosene and diesel** are obtained from petroleum.

✓ **Carbon and its Compounds as Fuels**

- © They release a large amount of heat and light on burning (high calorific value).
- © Most of them burn with a clean flame and leave little residue when there is enough air.
- © They are easily available, easy to store and transport, and their burning can be controlled.

✓ **Oxy-acetylene Flame for Welding**

- © A mixture of ethyne (acetylene) and pure oxygen, not air, is burnt for welding.
- © Ethyne burns completely in pure oxygen and produces a very high temperature (about 3000 °C), enough to melt metals.
- © In air, the oxygen is diluted by nitrogen, so ethyne burns incompletely with a sooty flame and a much lower temperature.

(ii) Oxidation

© Carbon compounds can be easily oxidised on combustion (this is **complete oxidation**).

© In addition to complete oxidation, there are reactions in which **alcohols are converted to carboxylic acids**:

$$CH_3CH_2OH \xrightarrow[\text{or Acidified } K_2Cr_2O_7 + \text{Heat}]{\text{Alkaline } KMnO_4 + \text{Heat}} CH_3COOH$$

© **Oxidising agents:** substances that are capable of adding oxygen to others. Alkaline potassium permanganate ($KMnO_4$) and acidified potassium dichromate ($K_2Cr_2O_7$) oxidise alcohols to acids, i.e. they add oxygen to the starting material – hence they are called oxidising agents.

© Conversion of ethanol (C_2H_6O) to ethanoic acid ($C_2H_4O_2$) is an **oxidation reaction** because oxygen is added to ethanol (and hydrogen is removed from it).

© **Alkaline potassium permanganate** = $KMnO_4$ + NaOH solution; **acidified potassium dichromate** = $K_2Cr_2O_7$ + H_2SO_4 solution. Both are strong oxidising agents.

© The oxygen supplied by the oxidising agent is **nascent oxygen** (freshly generated atomic oxygen, very reactive), written as **[O]** in the equation: $CH_3CH_2OH + 2[O] \rightarrow CH_3COOH + H_2O$.

© In fact, **all alcohols** can be oxidised to the corresponding carboxylic acids by strong oxidising agents.

✓ **Combustion and Oxidation – Difference**

© **Combustion:** the compound burns rapidly in the oxygen of air, breaks up completely into CO_2 and water vapour, and a large amount of heat and light is produced.

© **Oxidation:** a kind of **controlled combustion** – the compound combines with oxygen provided by an oxidising agent to form a new compound; it does not break down completely, and much less heat (if any) is produced.

✓ **Activity 4.5 – Oxidation of Ethanol**

(i) Procedure

- Take about 3 mL of ethanol in a test tube and warm it gently in a water bath.
- Add a 5% solution of alkaline potassium permanganate drop by drop.

(ii) Observation

- In the beginning, the pink/purple colour of potassium permanganate disappears.
- When excess potassium permanganate is added, its colour persists (does not disappear).

(iii) Reason

- At first, $KMnO_4$ is used up in oxidising ethanol to ethanoic acid, so its colour disappears.
- Once all the ethanol has been oxidised, the extra $KMnO_4$ is not used up, so its colour remains.

(iii) Addition Reaction

© **Unsaturated hydrocarbons add hydrogen** in the presence of catalysts such as **palladium or nickel** to give saturated hydrocarbons. Atoms such as halogens can also add across a double or triple bond.

© An **addition reaction** is a reaction in which an unsaturated hydrocarbon combines with another substance to give a **single product**. Hydrogen, chlorine (Cl_2) and bromine (Br_2) can all add to alkenes and alkynes.

© In the addition of hydrogen, one H atom adds to each carbon atom of the double bond, so the **double bond opens up into a single bond**.

© **Catalysts** are substances that cause a reaction to occur or proceed at a different rate without the reaction itself being affected (they remain chemically unchanged).

© General reaction:

$$R_2C = CR_2 + H_2 \xrightarrow{\text{Nickel catalyst}} R_2CH - CHR_2$$

© **Example:**

$$CH_2 = CH_2 + H_2 \xrightarrow{Ni} CH_3 - CH_3$$

(Ethene) (Ethane)

- © Adding hydrogen to an unsaturated compound is called **hydrogenation**. It is commonly used in the **hydrogenation of vegetable oils using a nickel catalyst** (to make vanaspati ghee).
- © **Conditions:** the oil is **heated with hydrogen** in the presence of **finely divided nickel** catalyst.
- © Vegetable oils such as **groundnut oil, cottonseed oil and mustard oil** contain double bonds and are **liquids** at room temperature. On hydrogenation they form saturated fats called **vegetable ghee (vanaspati ghee)**, which are **solid or semi-solid**.
- © Vegetable oils generally have **long unsaturated carbon chains**, while animal fats have **saturated carbon chains**.
- © Animal fats generally contain **saturated fatty acids**, which are said to be **harmful for health**. Oils containing **unsaturated fatty acids** should be chosen for cooking (that is why some vegetable oils are advertised as 'healthy').
- © Oils such as **sunflower oil, kardi (safflower) oil and groundnut oil** are good for health. Vanaspati ghee and animal fats like **butter and desi ghee** contain saturated fats and are said to be harmful if taken in large amounts.
- © Only unsaturated compounds undergo addition reactions. E.g., out of C_2H_6, C_3H_8, C_3H_6, C_2H_2 and CH_4, only **C_3H_6 (propene) and C_2H_2 (ethyne)** undergo addition reactions.

✓ **Test to distinguish Saturated and Unsaturated Hydrocarbons**

- © **Bromine water test:** a solution of bromine in water (bromine water) is **red-brown**. When it is added to an unsaturated compound, bromine adds across the double/triple bond and the red-brown colour is **discharged** (becomes colourless). A saturated compound does not decolourise bromine water. E.g. ethene and ethyne decolourise it; methane and ethane do not.
- © **Cooking oil vs butter:** add bromine water to each in separate test tubes – **cooking oil decolourises** bromine water (unsaturated), while **butter does not** (saturated).
- © **Flame test:** unsaturated hydrocarbons burn with a yellow, sooty flame; saturated hydrocarbons burn with a clean blue flame.

(iv) Substitution Reaction

- © Saturated hydrocarbons are fairly unreactive and are inert in the presence of most reagents.
- © However, in the presence of **sunlight**, chlorine reacts with hydrocarbons in a very fast reaction. Chlorine can **replace the hydrogen atoms one by one**.
- © It is called a **substitution reaction** because one type of atom or a group of atoms takes the place of another.
- © When the hydrogen atoms are replaced by chlorine, the reaction is also called **chlorination**.
- © Substitution reactions are a **characteristic property of saturated hydrocarbons (alkanes)** – methane, ethane, propane, butane, etc. Unsaturated hydrocarbons do not give substitution reactions with halogens; they give addition reactions.
- © **Example (Methane → Chloromethane):**

$$CH_4 + Cl_2 \xrightarrow{\text{Sunlight}} CH_3Cl + HCl$$

- © By supplying more chlorine, all the hydrogen atoms of methane can be replaced one by one, giving **dichloromethane** (methylene dichloride, CH_2Cl_2), **trichloromethane** (chloroform, $CHCl_3$ – once used as an anaesthetic) and **tetrachloromethane** (carbon tetrachloride, CCl_4). A number of products are usually formed with the higher homologues of alkanes.

❖ Some Important Carbon Compounds – Ethanol and Ethanoic Acid

- © Many carbon compounds are invaluable to us. Here we study the properties of two commercially important compounds – **ethanol** and **ethanoic acid**.

Ethanol (C_2H_5OH)

✓ **Properties of Ethanol**

- © Ethanol is a **liquid at room temperature** (melting point 156 K, boiling point 351 K).
- © Ethanol is the **second member** of the homologous series of alcohols (first member: methanol). Its formula is C_2H_5OH (CH_3CH_2OH); common name **ethyl alcohol**.
- © Ethanol is commonly called **alcohol** and is the active ingredient of all alcoholic drinks.
- © It is a **good solvent**, so it is used in medicines such as **tincture iodine, cough syrups and many tonics**.
- © Ethanol is **soluble in water in all proportions**.
- © It has a **burning taste**, is **lighter than water**, and is volatile with a low boiling point of **78 °C**. Its solubility in water is due to the **hydroxyl (–OH) group**.
- © Ethanol containing about 5% water (95% ethanol) is called **rectified spirit** (commercial alcohol). **100% pure ethanol** is called **absolute alcohol**.
- © Ethanol is a **covalent** compound with no hydrogen ions, so it is **neutral** and has no effect on litmus. All alcohols are neutral.
- © It is a colourless, volatile liquid with a characteristic pleasant smell; it is **neutral to litmus**.
- © Consumption of small quantities of dilute ethanol causes **drunkenness**. Even though this practice is condemned, it is socially widespread.
- © Intake of even a small quantity of **pure ethanol (absolute alcohol)** can be **lethal**. Long-term consumption of alcohol leads to many health problems.

✓ **Reactions of Ethanol**

(i) Combustion

- © Ethanol burns in air with a clean blue flame, giving carbon dioxide and water.
- © Ethanol is **highly inflammable** – it catches fire easily. All alcohols burn in air to give CO_2 and water. Because ethanol burns with a clean flame giving a lot of heat, it is used as a **fuel**.

$$\mathbf{CH_3CH_2OH + 3O_2 \rightarrow 2CO_2 + 3H_2O + heat\ and\ light}$$

(ii) Oxidation

- © Ethanol is oxidised to ethanoic acid by alkaline $KMnO_4$ or acidified $K_2Cr_2O_7$ on heating (see Oxidation above).

(iii) Reaction with Sodium

$$\mathbf{2Na + 2CH_3CH_2OH \rightarrow 2CH_3CH_2O^-Na^+ + H_2}$$

(Sodium ethoxide)

- © Alcohols react with sodium, leading to the **evolution of hydrogen**. With ethanol, the other product is **sodium ethoxide**.
- © This reaction is used as a **test for alcohols**.
- © Other substances that produce hydrogen on reacting with metals are **acids** (e.g., $Zn + 2HCl \rightarrow ZnCl_2 + H_2$).

✓ **Activity 4.6 – Reaction of Ethanol with Sodium (teacher's demonstration)**

(i) Procedure

- Drop a small piece of sodium, about the size of a couple of grains of rice, into ethanol (absolute alcohol).

(ii) Observation

- Sodium reacts, producing brisk effervescence (bubbles of a gas).

(iii) Test for the gas

- Bring a burning matchstick near the mouth of the test tube – the gas burns with a 'pop' sound. This shows the gas is hydrogen.

(iv) Reaction to give Unsaturated Hydrocarbon (Dehydration)

- © Heating ethanol at **443 K** with **excess concentrated sulphuric acid** results in the **dehydration** of ethanol to give **ethene**:

$$CH_3 - CH_2OH \xrightarrow[443\ K]{\text{Hot Conc. } H_2SO_4} CH_2 = CH_2 + H_2O$$

- © The concentrated sulphuric acid acts as a **dehydrating agent**, which removes water from ethanol.
- © **Dehydration** means removal of a water molecule. One H from the CH_3 group and the OH from the CH_2OH group are removed together as H_2O, forming a double bond (CH_2=CH_2). 443 K = 170 °C.

(v) Esterification

- © Ethanol reacts with ethanoic acid in the presence of an acid catalyst to form an ester (see Reactions of Ethanoic Acid below).

✓ **Tests for Alcohols**

- © **Sodium metal test:** add a small piece of sodium to the organic liquid in a dry test tube. **Effervescence of hydrogen gas** shows it is an alcohol (the gas burns with a 'pop' sound).
- © **Ester test:** warm the liquid with glacial ethanoic acid and a few drops of concentrated sulphuric acid. A **sweet (fruity) smell** of an ester shows it is an alcohol.

✓ **Uses of Ethanol**

- © In the manufacture of **paints, varnishes, medicines, perfumes, dyes, soaps and synthetic rubber**.
- © As a **solvent** – many organic compounds insoluble in water dissolve in ethanol; used in medicines such as **tincture iodine** (iodine dissolved in ethanol), cough syrups and tonics.
- © As a **fuel** in cars (along with petrol) and in **spirit lamps**.
- © In **alcoholic drinks** – whisky (about 35% ethanol), wine (10–20%) and beer (about 6%).
- © As an **antiseptic** to sterilise wounds and syringes in hospitals and dispensaries.

✓ **Effects of Alcohols on Living Beings**

- © When large quantities of ethanol are consumed, it **slows down metabolic processes** and **depresses the central nervous system**.
- © This results in lack of coordination, mental confusion, drowsiness, lowering of normal inhibitions and finally stupor. The person may feel relaxed without realising that judgement, sense of timing and muscular coordination are seriously impaired.
- © Unlike ethanol, intake of **methanol** in very small quantities can cause **death**. Methanol is oxidised to **methanal** in the liver; methanal reacts rapidly with the components of cells and **coagulates the protoplasm**, much like an egg is coagulated by cooking.
- © Methanol also affects the **optic nerve**, causing **blindness**.

✓ **Harmful Effects of Drinking Alcohol**

- © Alcohol slows down the nervous system and brain, so judgement is impaired and reactions become slow – **drunken driving** causes road accidents.
- © It lowers inhibitions, making a person quarrelsome – leading to **fights, violence and crime**.
- © Heavy drinking causes **staggering, slurred speech, blurred vision, dizziness and vomiting**; large quantities can cause unconsciousness and even death.
- © It makes a person **addicted (alcoholic)** and can make him financially bankrupt.
- © Long-term heavy drinking damages the **stomach, liver, heart and brain**; the liver disease **cirrhosis** can lead to death.
- © **Denatured alcohol:** ethanol is an important industrial solvent and is supplied **duty-free** (without tax) for industry, which makes it much cheaper. To prevent its misuse for drinking, industrial ethanol is made unfit for drinking by adding small amounts of poisonous substances like **methanol, pyridine or copper sulphate**. Copper sulphate (or dyes) give it a **blue colour** so that it can be identified easily.

✓ **Alcohol as a Fuel**

- © Sugarcane plants are one of the most efficient converters of sunlight into chemical energy.
- © Sugarcane juice is used to prepare **molasses**, which is **fermented** to give alcohol (ethanol).

© After sugar is crystallised from sugarcane juice, the dark liquid left behind is **molasses**, which still contains about **30% sugar**. This sugar is **fermented** (broken down by enzymes) to give ethanol.

© Some countries now use alcohol as an **additive in petrol** (e.g., gasohol / ethanol-blended petrol) since it is a **cleaner fuel** which gives only carbon dioxide and water on burning in sufficient air (oxygen).

© E.g. in **Brazil**, a mixture of ethanol and petrol (and even ethanol alone) is used as fuel for cars. Ethanol does not produce poisonous carbon monoxide, so adding it to petrol **reduces CO emission** from cars.

Ethanoic Acid (CH_3COOH)

✓ **Properties of Ethanoic Acid**

© Ethanoic acid is commonly called **acetic acid** and belongs to a group of acids called **carboxylic acids**.

© It is the **second member** of the homologous series of carboxylic acids (first member: methanoic acid). Its formula is CH_3COOH.

© A **5–8% solution** of acetic acid in water is called **vinegar**, which is used widely as a **preservative in pickles**.

© The melting point of pure ethanoic acid is **290 K** (17 °C), so it often **freezes during winter** in cold climates. This gave rise to its name **glacial acetic acid**. Its boiling point is 391 K.

© On cooling, pure ethanoic acid freezes to form a colourless, **ice-like solid that looks like a glacier** – hence the name glacial acetic acid.

© It is a colourless liquid with a sour taste and a pungent smell of vinegar; it is miscible with water.

© Carboxylic acids are characterised by their **acidic nature**. However, unlike mineral acids like HCl, which are **completely ionised**, carboxylic acids are **weak acids** (only partially ionised).

✓ **Activity 4.7 – Comparing Acetic Acid and Hydrochloric Acid**

(i) Procedure

- Compare the pH of dilute acetic acid and dilute hydrochloric acid using both litmus paper and universal indicator.

(ii) Observation

- Both acids turn blue litmus red, so both are indicated as acids by the litmus test.
- Universal indicator shows dilute HCl as red (pH about 1–2) and dilute acetic acid as orange-yellow (pH about 3–4).

(iii) Conclusion: Acetic acid is a weaker acid than hydrochloric acid – they are not equally strong.

✓ **Reactions of Ethanoic Acid**

(i) Esterification Reaction

© Esters are most commonly formed by the reaction of an **acid and an alcohol**. Ethanoic acid reacts with absolute ethanol in the presence of an **acid catalyst** to give an ester:

$$CH_3COOH + CH_3CH_2OH \xrightarrow[(\text{Conc. } H_2SO_4)]{\text{Acid}} CH_3COOCH_2CH_3 + H_2O$$

(Ethanoic acid) (Ethanol) (Ethyl ethanoate – an ester)

© Generally, esters are **sweet-smelling** substances. They are used in making **perfumes** and as **flavouring agents**.

© All carboxylic acids react with alcohols in the presence of a little concentrated sulphuric acid to form sweet-smelling esters. The formation of a sweet-smelling ester is used as a **test for alcohols as well as carboxylic acids**.

✓ **Activity 4.8 – Formation of an Ester**

(i) Procedure

- Take 1 mL ethanol (absolute alcohol) and 1 mL glacial acetic acid along with a few drops of concentrated sulphuric acid in a test tube.
- Warm in a water bath for at least five minutes.
- Pour into a beaker containing 20–50 mL of water and smell the resulting mixture.

(ii) Observation

- A sweet, fruity smell is produced.

(iii) Conclusion

- An ester (ethyl ethanoate) is formed. Concentrated sulphuric acid acts as the catalyst.

(ii) Saponification

© On treating with **sodium hydroxide** (an alkali), the ester is converted back to **alcohol** and the **sodium salt of the carboxylic acid**:

$$CH_3COOC_2H_5 \xrightarrow{NaOH} C_2H_5OH + CH_3COONa$$

© This reaction is known as **saponification** because it is used in the **preparation of soap**. Soaps are sodium or potassium salts of long-chain carboxylic acids.

© The **alkaline hydrolysis** (breaking down with an alkali) of esters is called saponification. When esters of higher fatty acids with glycerol (oils and fats) are hydrolysed with NaOH, sodium salts of higher fatty acids – **soaps** – are formed.

(iii) Reaction with a Base

© Like mineral acids, ethanoic acid reacts with a base such as sodium hydroxide to give a **salt** (sodium ethanoate, commonly called **sodium acetate**) and **water** (neutralisation):

$NaOH + CH_3COOH \rightarrow CH_3COONa + H_2O$

(iv) Reaction with Carbonates and Hydrogencarbonates

© Ethanoic acid reacts with carbonates and hydrogencarbonates to give a **salt, carbon dioxide and water**. The salt produced is commonly called sodium acetate.

© The reaction with sodium hydrogencarbonate (brisk effervescence of CO_2) is used as a **test for carboxylic acids**; all carboxylic acids decompose sodium hydrogencarbonate in this way.

$2CH_3COOH + Na_2CO_3 \rightarrow 2CH_3COONa + H_2O + CO_2$

$CH_3COOH + NaHCO_3 \rightarrow CH_3COONa + H_2O + CO_2$

✓ **Activity 4.9 – Reaction of Ethanoic Acid with Carbonates**

(i) Procedure

- Take a spatula full of sodium carbonate in a test tube and add 2 mL of dilute ethanoic acid.
- Pass the gas produced through freshly prepared lime water.
- Repeat the activity with sodium hydrogencarbonate instead of sodium carbonate.

(ii) Observation

- Brisk effervescence is seen in both cases.
- The gas turns lime water milky.

(iii) Conclusion

- The gas is carbon dioxide: $Ca(OH)_2 + CO_2 \rightarrow CaCO_3$ (milky) + H_2O.

✓ **Tests for Carboxylic Acids**

© **Sodium hydrogencarbonate test:** add a pinch of $NaHCO_3$ to the compound – **brisk effervescence of CO_2** (turns lime water milky) shows a carboxylic acid.

© **Litmus test:** a carboxylic acid turns **blue litmus red**.

© **Ester test:** warm the compound with ethanol and 2–3 drops of concentrated sulphuric acid – a **sweet smell** of an ester shows a carboxylic acid.

✓ **Uses of Ethanoic Acid**

© As **vinegar**, it is used as a food preservative in **pickles and sauces** (like tomato sauce) and for dressing food dishes (appetiser).

© For making **cellulose acetate**, an important artificial fibre.

© In the manufacture of **acetone (propanone)** and **esters** used in perfumes.

© In the preparation of **dyes, plastics and pharmaceuticals** (e.g., aspirin).

© To **coagulate rubber** from latex.

✓ **Distinguishing between Ethanol and Ethanoic Acid**

Property / Test	Ethanol	Ethanoic Acid
Melting point	156 K	290 K (freezes in cold winters)
Boiling point	351 K	391 K
Smell	Pleasant (alcoholic) smell	Pungent smell of vinegar
Taste	Burning taste	Sour taste
Effect on blue litmus	No change (neutral)	Turns red (acidic)
With $NaHCO_3$ / Na_2CO_3	No reaction	Brisk effervescence of CO_2
With NaOH	No reaction	Forms sodium ethanoate and water
With alkaline $KMnO_4$ on warming	Pink colour disappears (gets oxidised)	No change

© **Simplest test:** add sodium hydrogencarbonate – a carboxylic acid gives brisk effervescence of CO_2; an alcohol does not.

❖ SOAPS AND DETERGENTS

© Any substance that has a **cleansing action in water** (removes dirt) is called a **detergent** (cleansing agent). There are two types: **soapy detergents** (called **soaps**) and **non-soapy detergents** (called **synthetic detergents** or just detergents).

© Soaps and detergents are used for washing clothes, bathing, shaving, hair shampoos, cleaning utensils, etc.

Soaps

© Soaps are **sodium or potassium salts of long-chain carboxylic acids** (fatty acids), e.g. sodium stearate, $C_{17}H_{35}COONa$.

© Examples: **sodium stearate**, $C_{17}H_{35}COO^-Na^+$ (from stearic acid, $C_{17}H_{35}COOH$) and **sodium palmitate**, $C_{15}H_{31}COO^-Na^+$ (from palmitic acid, $C_{15}H_{31}COOH$).

© **Manufacture of soap:** fats and vegetable oils are esters of higher fatty acids with an alcohol called **glycerol**. Soap is made by heating animal fat or vegetable oil with concentrated sodium hydroxide solution (caustic soda) – this process is **saponification**.

Fat or Oil (ester) + Sodium hydroxide —Heat→ Soap (sodium salt of fatty acid) + Glycerol

© Oils used for making soap include **castor oil, cottonseed oil, soyabean oil, linseed oil, coconut oil, palm oil and olive oil**.

© Most dirt is **oily** in nature, and oil does not dissolve in water. Soap helps remove such dirt.

© Soap molecules have two different ends:

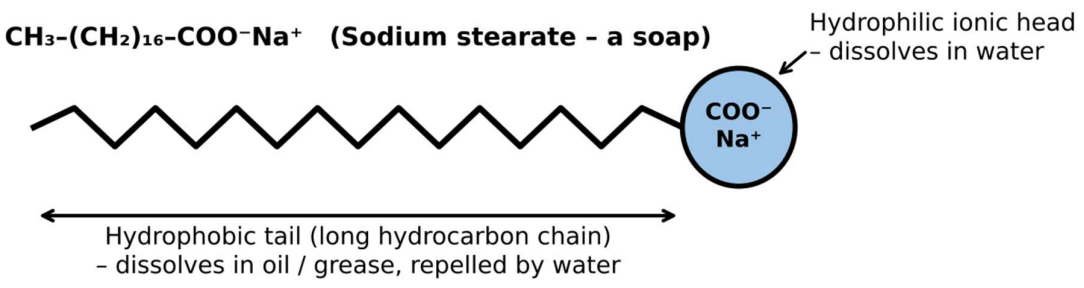
(i) Ionic end (hydrophilic):

© Interacts with water – attracted towards water and dissolves in water.

(ii) Carbon chain (hydrophobic):

© Interacts with oil and grease (hydrocarbons) – repelled by water.

© Because of its long 'tail' and small ionic 'head', a soap molecule is said to have a **tadpole structure**.



✓ **Activity 4.10 – Effect of Soap in Cleaning**

(i) Procedure

- Take about 10 mL of water each in two test tubes A and B.
- Add a drop of cooking oil to both test tubes.
- Add a few drops of soap solution to test tube B only.
- Shake both test tubes vigorously for the same period of time and then leave them undisturbed.

(ii) Observation

- Immediately after shaking, oil and water appear mixed in both test tubes.
- On standing, the oil layer separates out first (quickly) in test tube A; in test tube B the oil stays mixed for a long time.

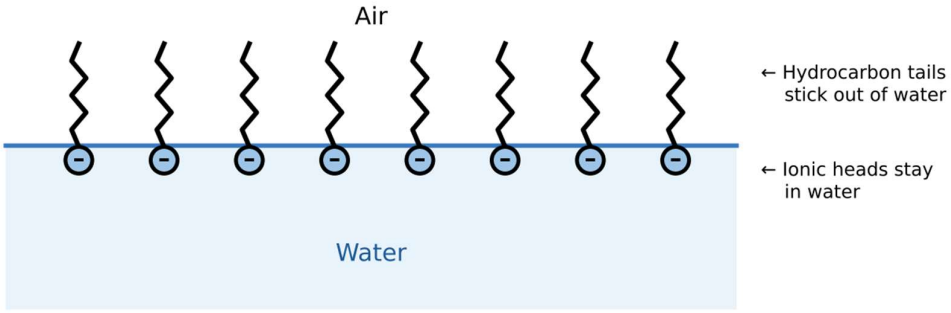
(iii) Conclusion

- Soap helps oil mix with water (forms an emulsion).
- This demonstrates the effect of soap in cleaning.

✓ **Orientation of Soap Molecules at Water Surface**

© When soap is at the surface of water, the hydrophobic 'tail' is not soluble in water.

© So soap molecules align along the surface of water with the **ionic end in water** and the **hydrocarbon 'tail' protruding out of water**.



✓ **Formation of Micelles**

© Inside water, soap molecules have a unique orientation that keeps the hydrocarbon portion out of the water.

© They form **clusters of molecules** in which the **hydrophobic tails are in the interior** of the cluster and the **ionic ends are on the surface**. This cluster is called a **micelle**.

© A micelle is a **spherical aggregate of soap molecules** in water. The ionic ends point outwards because their **negative charges repel** one another. A soap solution is a **colloidal solution**.

© In a micelle, one end of the molecules is towards the oil droplet while the ionic end faces outside. This forms an **emulsion** in water.

✓ **Cleaning Action of Soap**

© The ionic end of soap interacts with water while the carbon chain interacts with oil.

© The **oily dirt is collected in the centre of the micelle**.

© The micelles stay in solution as a **colloid** and do not come together to precipitate because of **ion–ion repulsion** (their outer surfaces carry the same charge).

© Thus the dirt suspended in the micelles is **easily rinsed away** with water, and clothes become clean.

© The soap micelles are large enough to **scatter light**, hence a **soap solution appears cloudy**.

✓ **Role of Agitation in Washing**

© Clothes are beaten on a stone, beaten with a paddle, scrubbed with a brush or agitated in a washing machine after adding soap.

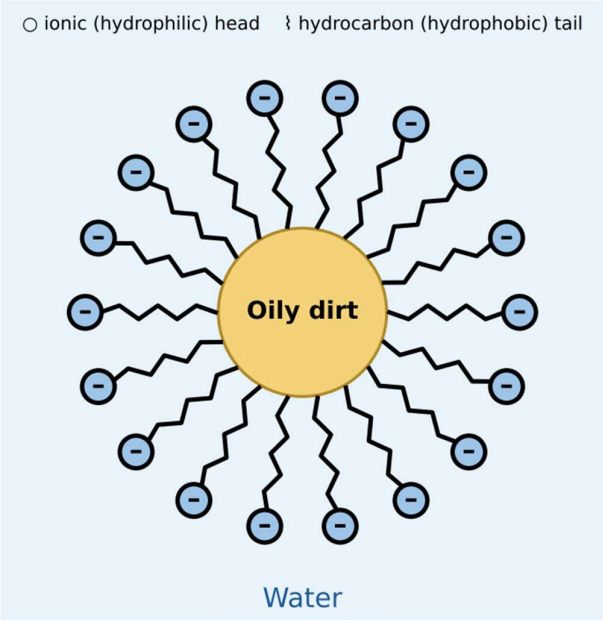
© Agitation helps soap molecules reach the oily dirt stuck on the fabric.

© It breaks the oily dirt into small droplets, which get trapped in micelles and are then easily rinsed away.

✓ **Micelle Formation in a Hydrocarbon**

© When soap is dissolved in a hydrocarbon (e.g., kerosene), the arrangement is reversed: the **ionic ends face inside** and the **hydrocarbon tails face outside**, because the hydrocarbon solvent attracts the hydrocarbon tails.

© Micelles do **not** form in ethanol, because soap dissolves completely in ethanol (both its ends are soluble), so there is no need for the molecules to cluster.



✓ **Problem of Using Soap in Hard Water**

© While bathing in hard water, **foam forms with difficulty** and an insoluble substance called **scum** remains after washing.

© Scum is formed by the reaction of soap with the **calcium and magnesium salts** that cause the hardness of water. E.g.:

$$\mathbf{2C_{17}H_{35}COONa + CaCl_2 \rightarrow (C_{17}H_{35}COO)_2Ca \downarrow (scum) + 2NaCl}$$

© Hence a **larger amount of soap** is needed to form lather in hard water, because part of the soap is wasted in forming scum.

© The scum (insoluble precipitate) **sticks to the clothes** being washed and interferes with the cleaning ability of the additional soap, making cleaning difficult.

© **Distilled water** is the softest; **rain** and **tap water** are soft. **Well water and hand-pump water** are hard.

✓ **Activity 4.11 – Effect of Soap on Soft and Hard Water**

(i) Procedure

- Take about 10 mL of distilled water (or rain water) and 10 mL of hard water (from a tubewell or hand-pump) in separate test tubes.
- Add a couple of drops of soap solution to both.
- Shake the test tubes vigorously for an equal period of time and observe the amount of foam formed.
- Note for the teacher: if hard water is not available, prepare it by dissolving hydrogencarbonates/sulphates/chlorides of calcium or magnesium in water.

(ii) Observation

- Foam is formed in distilled (soft) water but a white curdy precipitate (scum) is formed in hard water.

(iii) Reason: Hard water contains calcium and magnesium salts, which react with soap to form insoluble scum.

✓ **Nature of Soap**

© Soap is the salt of a **strong base** (sodium hydroxide) and a **weak acid** (carboxylic acid), so a soap solution is **alkaline (basic)**: it turns **red litmus blue** and has no effect on blue litmus.

Detergents

© The problem of soap in hard water is overcome by using another class of compounds called **detergents** as cleansing agents.

© Detergents are also called **'soap-less soaps'** – they act like soaps but do not contain the usual soaps like sodium stearate.

© Detergents are generally **sodium salts of long-chain benzene sulphonic acids** or **sodium salts of long-chain alkyl hydrogensulphates** (or ammonium salts with chloride or bromide ions). The ionic group is $\mathbf{-SO_3^-Na^+}$ or $\mathbf{-SO_4^-Na^+}$.

© Examples:

$\mathbf{CH_3–(CH_2)_{11}–C_6H_4–SO_3^-Na^+}$ **(Sodium n-dodecyl benzene sulphonate – a common detergent)**

$\mathbf{CH_3–(CH_2)_{10}–CH_2–SO_4^-Na^+}$ **(Sodium n-dodecyl sulphate)**

- © Detergents are made from long-chain **hydrocarbons obtained from petroleum**. Their cleansing action is similar to that of soap (through **micelle formation**).
- © Like soaps, they have a **long hydrocarbon chain** with a charged (ionic) end.
- © The **charged ends of detergents do not form insoluble precipitates** with the calcium and magnesium ions in hard water. Thus, they **remain effective in hard water**.
- © Detergents are usually used to make **shampoos** and products for **cleaning clothes** (washing powders).

✓ **Advantages and Disadvantage of Detergents**

- © Detergents can be used even with **hard water**, whereas soaps are not suitable for hard water.
- © Detergents have a **stronger cleansing action** than soaps; soaps have relatively weak cleansing action.
- © **Disadvantage:** some detergents are **non-biodegradable** – they cannot be decomposed by micro-organisms like bacteria, so they cause **water pollution** (foam) in lakes and rivers. Soaps are biodegradable and do not cause water pollution.

✓ **Activity 4.12 – Soap and Detergent in Hard Water**

(i) Procedure

- Take two test tubes with about 10 mL of hard water in each.
- Add five drops of soap solution to one and five drops of detergent solution to the other.
- Shake both test tubes for the same period.

(ii) Observation

- The test tube with detergent has much more foam.
- A curdy solid (scum) is formed in the test tube with soap.

(iii) Conclusion: Soap is not effective in hard water; detergents are effective in hard water.

✓ **Testing Hardness of Water**

- © A detergent forms lather easily in both hard and soft water, so it cannot be used to check whether water is hard.
- © Soap is used for this test: it forms scum and little lather in hard water, but plenty of lather in soft water.

✓ **Difference between Soaps and Detergents**

Basis	Soaps	Detergents
Chemical nature	Sodium/potassium salts of long-chain carboxylic acids	Sodium salts of sulphonic acids or ammonium salts with chloride/bromide ions
Ionic group	$-COO^-Na^+$	$-SO_3^-Na^+$ or $-SO_4^-Na^+$ (or $-N^+$ with Cl^-/Br^-)
In hard water	Form scum; not effective	No scum; effective
Amount needed in hard water	Large amount	Small amount
Raw material	Vegetable oils / animal fats	Mainly petroleum products
Cleansing action	Relatively weak	Strong
Biodegradability	Biodegradable	Many are non-biodegradable and cause water pollution
Uses	Bathing soaps, washing soaps	Shampoos, washing powders, cleaning products

✓ **Group Activity – Preparation of Soap**

- © Take about 20 mL of castor oil / cotton seed oil / linseed oil / soyabean oil in a beaker and add 30 mL of 20% sodium hydroxide solution.
- © Heat the mixture with continuous stirring for a few minutes till it thickens (saponification).
- © Add 5–10 g of common salt, stir well and allow it to cool. The salt helps the soap to separate out (salting out).
- © Most soap separates on its own, but some remains dissolved. Common salt **decreases the solubility of soap** in water, so all the soap separates out (precipitates) as a solid.
- © The soap can be cut into fancy shapes; perfume can be added before it sets.

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